



# BOOST YOUR VOCABULARY

• Second Edition •

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academic words from

**CAMBRIDGE IELTS 11**

# TEST 1

## READING PASSAGE 1



**B**y the year 2050, nearly 80% of the Earth's population will live in **urban** centres. Applying the most **conservative** estimates to **current demographic** trends, the human population will increase by about three billion people by then. An estimated 109 hectares of new land (about 20% larger than Brazil) will be needed to grow enough food to feed them, if traditional farming methods continue as they are practised today. At present, throughout the world, over 80% of the land that is suitable for **raising crops** is in use. Historically, some 15% of that has been laid waste by poor management **practices**. What can be done to **ensure** enough food for the world's population to live on? The concept of indoor farming is not new, since **hothouse** production of tomatoes and other produce

**Urban**= city, inner-city, metropolitan, town # rural, suburban (*urb=city. i.e. suburb, urbanity*) '3:bən  
**Conservative**=be likely to be less than the real amount. kən's3:vətɪv  
**Current**= present, existing, recent. 'kʌrənt  
**Demographic**= relating to the population and groups of people in it (*dem=people, i.e: democracy*) demə'græfɪk  
**raise crops** = grow plants such as wheat, rice, or fruits... reɪz krops  
**Practice**= habit, tradition, or custom 'præktɪs  
**Ensure**= make sure, confirm, make certain. ɪn'ʃʊə  
**Hothouse** = a heated glass building in which plants are grown. 'hoʊhaʊs

has been in **vogue** for some time. What is new is the **urgent** need to **scale up** this technology to **accommodate** another three billion people. Many believe an entirely new approach to indoor farming is required, **employing cutting-edge** technologies. One such **proposal** is for the 'Vertical Farm'. The concept is of **multi-storey** buildings in which food crops are grown in environmentally controlled conditions. Situated in the heart of urban centres, they would **drastically** reduce the amount of transportation required to bring food to consumers. Vertical farms would need to be efficient, cheap to construct and safe to operate. If successfully **implemented**, **proponents** claim, vertical farms offer the promise of **urban renewal**, **sustainable** production of a safe and varied food supply (through **year-round** production of all crops), and the eventual repair of ecosystems that have been **sacrificed** for **horizontal farming**.

It took humans 10,000 years to learn how to grow most of the crops we now **take for granted**. Along the way, we **despoiled** most of the land we worked, often turning **verdant**, natural ecozones into **semi-arid** deserts. Within that same time frame, we **evolved into** an urban species, in which 60% of the human population now lives vertically in cities. This means that, for the majority, we humans have **shelter** from the elements, yet we **subject** our **food-bearing plants** to the **rigours** of the great outdoors and can do no more than hope for a good weather year. However, more often than not now, due to a rapidly changing climate, that is not what

**Vogue**= fashion, trend, rage/fad. vʊg  
**Urgent**= very important and needing to be dealt with immediately. 'ɜ:dʒənt  
**Scale up**= increase, expand, develop. skeɪl ʌp  
**Accommodate**= to give someone what is needed. ə'kɒmədeɪt  
**Employ**= use, utilize, exploit, apply. ɪm'plɔɪ  
**Cutting-edge**= modern, newest, most advanced. 'kʌtɪŋ edʒ  
**Proposal**= suggestion, application, plan. prə'pəʊzəl  
**Vertical farming**= an idea for a way of farming in which plants are grown or animals are kept in tall structures with many levels. 'vɜ:tɪkl 'fɑ:mɪŋ  
**Multi-storey**= many floors. 'mʌlti - 'sto:ri  
**Drastically**= radically, extremely, significantly. 'dræstɪklɪ  
**Implement**= fulfill, put into practice, take action. 'ɪmpləment  
**Proponent**= advocate, supporter >< opponent. prə'pəʊnənt  
**Urban renewal**= the improvement and sometimes replacement of buildings in a city, (re-= do again, i.e: rebuild, reusable..) 'ɜ:bən rɪ'nju:əl  
**Sustainable**= environmentally friendly, eco-friendly, green. sə'steɪnəbəl  
**Year-round**= happening or continuing through the whole year. 'jiə - 'raʊnd  
**Sacrifice**= decide not to have something in order to get something that is more important 'sækrɪfaɪs  
**Horizontal farming**= traditional farming, crops are planted at only one level >< vertical farming. hɒrɪ'zɒntl 'fɑ:mɪŋ  
  
**Take sb/st for granted**= do not realize or show that sb/st is greatful. teɪk 'sʌmbədi 'sʌmθɪŋ fə 'grɑ:ntɪd  
**Despoil**= damage, spoil, ruin (de-= reduce, i.e decline) dɪ'spɔɪl  
**Verdant**= lush, green, grassy 'vɜ:dənt  
**Semi-arid**= having little rain but not completely dry (Semi-= half, i.e: semi-natural) 'semi - 'æɪrɪd  
**Evolve**= develop gradually, change, grow {vol- = turn i.e: revolt, evolution} ɪ'vɒlv  
**Shelter**= cover, protection from danger. 'ʃeltə  
**Subject smt to smt**= cause smt to experience smt, especially smt unpleasant sʌb'dʒekt 'sʌmθɪŋ tu 'sʌmθɪŋ  
**Food-bearing plants**= plants provide food such as tomato, vegetable. fu:d - 'beərɪŋ plɑ:nts  
**The rigours of something**= the unpleasant or severe conditions of smth. ðə 'rɪgəz əv 'sʌmθɪŋ

happens. Massive floods, long **droughts**, hurricanes and severe **monsoons take their toll** each year, destroying millions of tons of valuable crops.

The supporters of vertical farming claim many potential advantages for the system. For instance, crops would be produced all year round, as they would be kept in artificially controlled, **optimum** growing conditions.

There would be no weather-related crop **failures** due to droughts, floods or **pests**. All the food could be grown organically, **eliminating** the need for **herbicides**, **pesticides** and **fertilisers**. The system would greatly reduce the **incidence** of many **infectious** diseases that are acquired at the agricultural **interface**. Although the system would consume energy, it would return energy to the grid via methane generation from **composting** non **edible** parts of plants. It would also dramatically reduce fossil fuel use, by cutting out the need for **tractors**, **ploughs** and shipping.

A major **drawback** of vertical farming, however, is that the plants would require **artificial** light. Without it, those plants nearest the windows would be **exposed** to more sunlight and grow more quickly, reducing the efficiency of the system. Single-storey greenhouses have the benefit of natural overhead light; even so, many still need artificial lighting.

A multi-storey facility with no natural overhead light would require far more. Generating enough light could be **prohibitively** expensive, unless cheap, renewable energy is available, and this appears to be rather a future **aspiration** than a **likelihood** for the near future.

**Drought**= a long period of dry weather when there is not enough water *draʊt*

**Hurricane**= cyclone, typhoon, tornado, storm *ˈhʌrɪkən*

**Monsoon**= heavy rain, rains. *mɒnˈsuːn*

**Take their toll**= to have a very bad effect on smt or smb over a long period of time *ˈteɪk ðəə təʊl*

**Optimum**= most favorable, best, greatest (*optim*= best, i.e: **optimal**) *ˈoptɪməəm*

**Failure** = not successful. *ˈfeɪljə*

**Pest**= an insect or small animal that is harmful or damages crops *pest*

**Eliminate** = remove, eradicate, put an end to, get rid of. *ɪˈlɪmənət*

**Herbicide**= a substance used to kill unwanted plants (*herb*=grass, i.e **herbal**, **herbivore** # *cid*=kill i.e **pesticide**) *ˈhɜːbɪsaɪd*

**Pesticide**= a chemical substance used to kill insects and small animals that destroy crops(*cid*=kill) *ˈpestɪsaɪd*

**Fertilizer**= a substance that is put on the soil to make plants grow *ˈfɜːtəlaɪzə*

**Incidence**= occurrence, frequency, rate. *ˈɪnsɪdəns*

**Infectious**=, able to pass a disease from one person, animal, or plant to another *ɪnˈfekʃəs*

**Interface**=edge, border, line (*inter* = among, between, i.e: **intersection**) *ˈɪntəfeɪs*

**Compost**= to collect and store plant material so it can decay and be added to soil to improve its quality. *ˈkɒmpɒst*

**Edible**=suitable to be eaten, not poisonous (*ed*=eat) *ˈedəbəl*

**Tractor**= a strong vehicle with large wheels, used for pulling farm machinery (*tract*=pull, draw. i.e: **traction**) *ˈtræktə*

**Plough**= a piece of farm equipment used to turn over the earth so that seeds can be planted *plau*

**Drawback**= disadvantage, downside, negative. *ˈdrɔːbæk*

**Artificial**= synthetic, non-natural, man-made *ɑːtɪˈfɪʃəl*

**Expose**= show, reveal, display (*ex*= out, i.e: **exterior**, **exclude**) *ɪkˈspəʊz*

**Prohibitively**= at a very high price that does not seem reasonable. *prəˈhɪbətɪv*

**Aspiration**= ambition, goal, aim, target *æspəˈreɪʃən*

**Likelihood**= probability, possibility *ˈlaɪklihud*

One **variation** on vertical farming that has been developed is to grow plants in **stacked trays** that move on rails. Moving the trays allows the plants to get enough sunlight. This system is already in operation, and works well within a single-storey greenhouse with light reaching it from above: it is not certain, however, that it can be made to work without that **overhead** natural light.

Vertical farming is an attempt to **address** the undoubted problems that we face in producing enough food for a growing population. At the moment, though, more needs to be done to reduce the **detrimental** impact it would have on the environment, particularly as regards the use of energy. While it is possible that much of our food will be grown in **skyscrapers** in future, most experts currently believe it is far more likely that we will simply use the space available on urban rooftops.

**Variation** = difference (**vari**= vary, i.e: **variety**) *veəri'eɪʃən*

**Stacked trays** = Having sections that are arranged vertically. *stækt treɪz*

**Overhead** = above, upstairs >< below *əʊvə'hed*

**Address**= tackle, deal with *ə'dres*

**Detrimental**= harmful, damaging, negative *detrə'mentl*

**Skyscraper**= a very tall modern city building *'skaɪskreɪpə*

## READING PASSAGE 2



**A** unique engineering achievement

The Falkirk Wheel in Scotland is the world's first and only **rotating** boat lift. Opened in 2002, it is **central** to the ambitious £84.5m Millennium Link project to **restore** navigability across Scotland by reconnecting the historic waterways of the Forth & Clyde and Union Canals.

The major challenge of the project lays in the fact that the Forth & Clyde Canal is situated 35 metres below the level of the Union Canal. Historically, the two canals had been joined near the town of Falkirk by a **sequence** of 11 locks - **enclosed** sections of canal in which the water level could be raised or lowered - that stepped down across a distance of 1.5 km. This had been **dismantled** in 1933, thereby breaking the link. When the project was **launched** in 1994, the British Waterways **authority** were keen to create a dramatic twenty-first-century **landmark** which would not only be

**Rotating**= turning in a circle, especially around a fixed point *rəʊˈteɪɪŋ*

**Central**= vital, essential, chief, most important, crucial, significant *ˈsentrəl*

**Restore**= Repair, rebuild *riˈstɔː*

**Navigability**= the degree to which an area of water is deep, wide, or safe enough for a boat to go through *ˈnævəgəbəl*

**Sequence**=chain, series (**sequ**=follow .i.e **consequence**, **subsequent**) *ˈsiːkwəns*

**Enclosed**=surrounded by walls, objects, or structures *ɪnˈkləʊz*

**Dismantle**= take to pieces, take apart *dɪsˈmæntl*

**Launch**= start *lɔːntʃ*

**Authority**= government department *ɔːˈθɒrəti*

**Landmark**= a building or place that is easily recognized *ˈlændmɑːk*

a **fitting commemoration** of the Millennium, but also a lasting symbol of the economic regeneration of the region.

Numerous ideas were **submitted** for the project, including concepts ranging from rolling eggs to tilting tanks, from **giant seesaws** to overhead **monorails**.

The eventual winner was a plan for the huge rotating steel boat lift which was to become The Falkirk Wheel.

The unique shape of the structure is claimed to have been inspired by various sources, both manmade and natural, most **notably** a Celtic double headed **axe**, but also the vast turning **propeller** of a ship, the **ribcage** of a whale or the **spine** of a fish.

The various parts of The Falkirk Wheel were all constructed and **assembled**, like one giant toy building set, at Butterley Engineering's Steelworks in Derbyshire, some 400 km from Falkirk. A team there carefully assembled the 1,200 tonnes of steel, **painstakingly** fitting the pieces together to an accuracy of just 10 mm to ensure a perfect final fit. In the summer of 2001, the structure was then dismantled and transported on 35 **lorries** to Falkirk, before all being **bolted** back together again on the ground, and finally lifted into position in five large sections by **crane**. The Wheel would need to **withstand immense** and constantly changing stresses as it rotated, so to make the structure more **robust**, the steel sections

**Fitting**= suitable or right for a particular occasion. 'fɪtɪŋ

**Commemoration**= something that makes you remember and respect someone important or an important event in the past.

(**memor**=remember .i.e **memorable**, **memory**)  
kəmemə'reɪʃən

**Submit**= offer, propose, suggest(**mit**=send .i.e **emit**, **transmit**) səb'mɪt

**Giant**= extremely large 'dʒaɪənt

**Seesaw**= a piece of equipment that children play on, made of a board that is balanced in the middle, so that when one end goes up the other goes down 'si:sɔ:

**Monorail**= a railway system that uses a single rail, usually high above the ground

(**mono**= one, single, ig: **monopoly**, **monolingual**) 'mɒnəʊreɪl

**Notably**= especially, most important 'nəʊtəbli

**Axe**=a tool that has a heavy iron or steel blade at the end of a long wooden handle, used for cutting wood æks

**Propeller**= a piece of equipment consisting of two or more blades that spin around, which makes an aircraft or ship move prə'pelə

**Ribcage**=the structure of ribs that protects your heart and lungs in your chest 'rɪbkɛɪdʒ

**Spine**=the line of bones down the centre of the back that provides support for the body spain

**Assemble**= bring together, put together, gather ə'sembəl

**Painstakingly** =carefully 'peɪnzteɪkɪŋli

**Lorry**= a large vehicle for carrying heavy goods 'lɒri

**Bolt**= fasten something with a metal pin or bar bæʊlt

**Crane**= hoist (a large tall machine used by builders for lifting heavy things) kreɪn

**Withstand**= resist, stand up to= to be strong enough to remain unharmed by something such as great heat, cold, pressure, etc.

wɪð'stænd

**Immense**= extremely large, enormous ɪ'mens

**Robust**= strong, tough rə'bʌst

were bolted rather than **welded** together. Over 45,000 bolt holes were matched with their bolts, and each bolt was hand-**tightened**.

The Wheel **consists of** two sets of opposing axe-shaped arms, **attached** about 25 metres apart to a fixed central spine. Two **diametrically** opposed water-filled '**gondolas**', each with a **capacity** of 360,000 litres, are fitted between the ends of the arms. These gondolas always weigh the same, whether or not they are carrying boats. This is because, according to Archimedes' principle of displacement, **floating** objects **displace** their own weight in water. So when a boat enters a gondola, the amount of water leaving the gondola weighs exactly the same as the boat. This keeps the Wheel balanced and so, despite its **enormous** mass, it rotates through 180° in five and a half minutes while using very little power. It takes just 1.5 kilowatt-hours (5.4 MJ) of energy to rotate the Wheel -roughly the same as boiling eight small **domestic kettles** of water.

Boats needing to be lifted up enter the **canal basin** at the level of the Forth & Clyde Canal and then enter the lower gondola of the Wheel. Two **hydraulic** steel gates are raised, so as to **seal** the gondola off from the water in the canal basin. The water between the gates is then **pumped out**. A hydraulic **clamp**, which prevents the arms of the Wheel moving while the gondola is docked, is removed, allowing the Wheel to turn. In the central machine room an **array** of ten hydraulic motors then begins to rotate the central **axle**. The axle connects to the outer arms of the Wheel, which begin to rotate at a speed of 1/8 of a **revolution** per minute. As the wheel rotates, the gondolas are kept in the upright position by a simple gearing system. Two eight-metre-wide **cogs orbit** a fixed inner cog of the same width, connected by two smaller cogs travelling in the opposite direction to the outer cogs -

**Weld**= to join pieces of metal together by heating *weld*  
**Tightened**= make something less easy to move *'taɪtnd*

**Consist of sth**= to be formed from the people or things mentioned *kən'sɪst əv 'sʌmθɪŋ*

**Attach**= to fasten or connect one object to another *ə'tætʃ*

**Diametrically**= completely *də'reɪ'metrɪkli*

**Gondola**= a long narrow boat with a flat bottom and high points at each end, used on the canals in Venice in Italy *'gɒndələ*

**Capacity**=the number of things or people that a container or space can hold *kə'pæsəti*

**Float**= to stay on the surface of a liquid and not sink *fləʊt*

**Displace**= put out of place, move *dis'pleɪs*

**Enormous**=extremely large or great *i'nɔ:məs*

**Domestic**= relating to the home or family *də'mestɪk*

**Kettles**= a container with a lid, handle and a spout, used for boiling water *'ketl*

**Canal**= a channel of water *kə'næl*

**Basin**= an area of land around a large river with streams running down into it *'beɪsən*

**Hydraulic**= moved or operated by the pressure of water or other liquid (*hydr*=water .i.e **hydrogen**, **hydrogen**) *haɪ'drɒlɪk*

**Seal**= shut out, close up, stop entering *si:l*

**Pumped out**=to remove water or other liquid from something using a pump: *pʌmpt 'aʊt*

**Clamp**= a piece of equipment for holding things together *klæmp*

**Array**= a large group of things is shown in an attractive way *ə'reɪ*

**Axle**= a bar connected to the centre of a circular object such as a wheel that allows or causes it to turn, especially one connecting two wheels of a vehicle *'æksəl*

**Revolution**= a complete circular movement around a point *revə'lʊ:ʃən*

**Cog**= a wheel with small bits sticking out around the edge that fit together with the bits of another wheel as they turn in a machine *kɒg*

**Orbit**= to move in a curved path around a much larger object *'ɔ:bit*

so ensuring that the gondolas always remain level. When the gondola reaches the top, the boat passes straight onto the **aqueduct** situated 24 metres above the canal basin.

The remaining 11 metres of lift needed to reach the Union Canal is achieved by means of a pair of locks. The Wheel could not be constructed to **elevate** boats over the full 35-metre difference between the two canals, owing to **the presence of** the historically important Antonine Wall, which was built by the Romans in the second century AD. Boats travel under this wall via a **tunnel**, then through the locks, and finally on to the Union Canal.

**Aqueduct**= a structure like a bridge, that carries water across a river or valley 'ækwədʌkt

**Elevate**= raise, lift, make higher(lev=rise.i.e *elevation, elevator*) 'elɪveɪt

**The presence of something**= the fact that someone or something is in a place ðə 'preznəs 'sʌmθɪŋ

**Tunnel**= a passage that has been dug under the ground for cars, trains etc to go through 'tʌnl

## READING PASSAGE 3



**M**ark Rowe reports on the increasingly **ambitious** **geo-engineering** projects being explored by scientists

A

Such is our **dependence** on **fossil fuels**, and such is the **volume** of carbon dioxide already released into the **atmosphere**, that many experts agree that significant global warming is now **inevitable**. They believe that the best we can do is keep it at a **reasonable** level, and at present the only serious option for doing this is cutting back on our carbon **emissions**. But while a few countries are making major **strides** in this regard, the majority are having great difficulty even **stemming** the rate of increase, let alone **reversing** it. Consequently, an increasing number of scientists are beginning to

**Geo-engineering**= the study of finding ways to change the earth's atmosphere in order to reduce global warming *dʒi:əʊendʒe'niəriŋ*

**Ambitious**= determined to be successful, rich, powerful, etc. *æm'biʃəs*

**Dependence on**= the situation in which you need something all the time to continue existing *di'pendəns on*

**Fossil fuel**= a fuel such as coal or oil that is produced by the very gradual decaying of animals or plants over millions of years *'fosl'fju:əl*

**Volume**=capacity, size, extent *'vɒlju:m*

**Atmosphere**= air in environment *'ætməsfɪə*

**Inevitable**= unavoidable, certain *ɪ'nevətəbəl*

**Reasonable** = acceptable, appropriate *'ri:zənəbəl*

**Emission**= release, discharge *ɪ'mɪʃən*

**Stride**= advance, progress, development *straɪd*

**Stem**= stop *stem*

**Reverse**= to change something, such as a decision, judgment, or process so that it is the opposite of what it was before (**re**=again, back.i.e **react, recede**) *ɪ'vɜ:s*

explore the **alternative** of geo-engineering — a term which generally refers to the **intentional** large-scale **manipulation** of the environment. According to its **proponents**, geo-engineering is the **equivalent** of a **backup** generator: if Plan A - reducing our dependency on fossil fuels - fails, we require a Plan B, employing **grand schemes** to slow down or reverse the process of global warming.

B

Geo-engineering; has been shown to work, at least on a small **localised** scale. For decades, MayDay **parades** in Moscow have taken place under clear blue skies, **aircraft** having **deposited** dry ice, silver iodide and cement powder to **disperse** clouds. Many of the schemes now suggested look to do the opposite, and reduce the amount of sunlight reaching the planet. The most **eye-catching** idea of all is suggested by Professor Roger Angel of the University of Arizona. His scheme would **employ** up to 16 trillion **minute** spacecraft, each weighing about one gram, to form a **transparent**, sunlight-refracting sunshade in an **orbit** 1.5 million km above the Earth. This could, argues Angel, reduce the amount of light reaching the Earth by two per cent.

C

The majority of geo-engineering projects so far carried out — which include planting forests in deserts and **depositing** iron in the ocean to **stimulate** the growth of algae - have focused on achieving a general cooling of the Earth. But some look specifically at **reversing** the melting at the poles, particularly the Arctic. The reasoning is that if you **replenish** the ice sheets and frozen waters of the high latitudes, more light will be **reflected** back into space, so reducing the warming of the oceans and atmosphere.

D

The concept of releasing **aerosol** **sprays** into the **stratosphere** above the Arctic has been **proposed** by several scientists. This would involve using sulphur or hydrogen sulphide aerosols so that sulphur dioxide

**Alternative**= an alternative idea, plan etc is different from the one you have and can be used instead ɒ.l'tɜːnətɪv

**Intentional**= planned, intended ɪn'tenʃənəl

**Manipulation**= the action of influencing or controlling something mənɪpjuː'leɪʃən

**Proponent**= advocate, supporter prə'pəʊnənt

**Equivalent**= parallel, similar (**equ**=equal.i.e **equate**, **equilibrium**) ɪ'kwɪvələnt

**Backup**= smt that you can use to replace something that does not work or is lost 'bækʌp

**Grand scheme**= important and large plan 'grænd ski:m

**Localize**= to limit smt to a particular area 'ləʊkəlaɪz

**Parade**= a public celebration when musical bands, brightly decorated vehicles etc move down the street pə'reɪd

**Aircraft**= a plane or other vehicle that can fly 'eəkrɑːft

**Deposit**= place, drop, put down dɪ'pɒzɪt

**Disperse**= melt away, get rid of dɪ'spɜːs

**Eye-catching**= attractive, noticeable 'aɪ kætʃɪŋ

**Employ**= use ɪm'plɔɪ

**Minute**= tiny, little, small 'mɪnət

**Transparent**= see-through, clear træn'spærənt

**Orbit**= a curved path followed by a planet or an object as it moves around another planet, star, moon, etc. 'ɔːbɪt

**Deposit**= put, place, lay dɪ'pɒzɪt

**Stimulate**= quicken, speed up, promote 'stɪmjələɪt

**Reverse**= to change something, such as the direction, order, or process, so that it is the opposite of what it was before rɪ'vɜːs

**Replenish**= refill (**plen**=full .i.e **plenty**) rɪ'plenɪʃ

**Reflect**= to show the image of smb/smt on the surface of smt (**fle**=bend.i.e **reflex**, **flexible**) rɪ'flekt

**Aerosol**= a metal container in which liquids are kept under pressure and forced out in a spray 'eəresəl

**Spray**= liquid which is forced out of a special container in a stream of very small drops spreɪ

**Stratosphere**= a very high position 'strætəsfiə

**Propose**= suggest, recommend prə'pəʊz

would form clouds, which would, in turn, lead to a global **dimming**. The idea is modelled on historic volcanic **explosions**, such as that of Mount Pinatubo in the Philippines in 1991, which led to a short-term cooling of global temperatures by 0.5 °C. Scientists have also **scrutinised** whether it's possible to **preserve** the ice sheets of Greenland with **reinforced high-tension** cables, preventing icebergs from moving into the sea. Meanwhile in the Russian Arctic, geo-engineering plans include the planting of millions of birch trees. Whereas the regions native **evergreen** pines shade the snow and absorb radiation, birches would **shed** their leaves in winter, thus enabling radiation to be reflected by the snow. **Re-routing** Russian rivers to increase cold water flow to ice-forming areas could also be used to slow down warming, say some climate scientists.

E

But will such schemes ever be **implemented**? Generally speaking, those who are most **cautious** about geo-engineering are the scientists involved in the research. Angel says that his plan is 'no **substitute** for developing renewable energy: the only **permanent** solution'. And Dr Phil Rasch of the US-based Pacific Northwest National Laboratory is equally **guarded** about the role of geo-engineering: 'I think all of us agree that if we were to end geo-engineering on a given day, then the planet would return to its pre-engineered condition very rapidly, and probably within ten to twenty years. That's certainly something to worry about.'

F

The US National Center for Atmospheric Research has already suggested that the proposal to **inject** sulphur into the atmosphere might affect rainfall patterns across **the tropics** and the Southern Ocean. 'Geo-engineering plans to inject stratospheric aerosols or to seed clouds would act to cool the planet, and act to increase the extent of sea ice,' says Rasch. 'But all the models suggest some impact on the **distribution** of **precipitation**.'

G

A further risk with geo-engineering projects is that you

**Dim**= make less bright, make less intense  
dim

**Explosion**= bang ɪk'spləʊʒən

**Scrutinise**= examine, study, analyze  
'skru:tənaɪz

**Preserve**= protect (serv=protect .i.e conserve, reserve) prɪ'zɜ:v

**Reinforce**= strengthen, support ri:ən'fɔ:s

**High-tension**= strong, tight haɪ 'tenʃn

**Evergreen**= an evergreen tree or bush does not lose its leaves in winter 'evəgrɪn

**Absorb**= take in or soak up (energy, liquid or other substance...) əb'sɔ:b

**Radiation**= powerful and very dangerous rays come especially from nuclear reactions reɪdɪ'eɪʃən

**Shed**= lose, get rid of, drop ʃed

**Enable**= make it possible for someone / something ɪ'neɪbəl

**Re-routing**= change the direction ri:'ru:tɪŋ

**Implement**= put into practice, apply  
'ɪmpləment

**Cautious**= careful 'kɔ:ʃəs

**Substitute**= alternate, replacement  
'sʌbstɪtju:t

**Renewable energy**= type of energy replaces itself naturally or is easily replaced because there is a large supply of it rɪ'nju:əbl  
'enədʒi

**Permanent**= everlasting, eternal, enduring  
'pɜ:mənənt

**Guard**= protect, defend ɡɑ:d

**Inject**= insert, add, bring in ɪn'dʒekt

**The tropics**= the hottest part of the world, which is around the equator ðə 'trɒpɪks

**Distribution**= spreading, allocation  
dɪstrə'bju:ʃən

**Precipitation**= rainfall prɪsɪpɪ'teɪʃən

can “**overshoot**” says Dr Dan Hunt, from the University of Bristol’s School of Geophysical Sciences, who has studied the likely **impacts** of the sunshade and aerosol schemes on the climate. ‘You may bring global temperatures back to pre-industrial levels, but the risk is that the **poles** will still be warmer than they should be and the tropics will be cooler than before **industrialisation**. “To avoid such a **scenario**,” Hunt says, “Angel’s project would have to operate at half strength; all of which reinforces his view that the best option is to avoid the need for geo-engineering altogether.”

H

The main reason why geo-engineering is supported by many in the scientific community is that most researchers have little **faith** in the ability of politicians to agree - and then bring in - the necessary carbon cuts. Even leading **conservation** organisations see the value of investigating the potential of geo-engineering. According to Dr Martin Sommerkorn, climate change advisor for the World Wildlife **Fund**’s International Arctic Programme, ‘**Human-induced** climate change has brought humanity to a position where we shouldn’t **exclude** thinking thoroughly about this topic and its possibilities.’

**Overshoot**= exceed, surpass əʊvə'ʃu:t

**Impact** = influence, effect 'impækt

**Pole**= the most northern or most southern point on a planet, especially the Earth pəʊl

**Industrialisation**= the process of developing industries in a country or an area ɪndʌ'striəlaɪ'zeɪʃən

**Scenario**= the description of possible actions or events in the future sə'na:riəʊ

**Faith**= belief feɪθ

**Conservation**= Preservation, protection kən'sə'veɪʃən

**Potential**= likely to develop into a particular type of person or thing in the future pə'tenʃəl

**Human-induced**= caused by human 'hju:mən - ɪn'dʒu:st

**Exclude**= omit, miss out, eliminate, not include ɪk'sklu:d

## Test 2

### READING PASSAGE 1



# Raising the Mary Rose

**H**ow a sixteenth-century warship was recovered from the seabed.

On 19 July 1545, English and French **fleets** were **engaged in** a sea battle off the coast of southern England in the area of water called the Solent, between Portsmouth and the Isle of Wight. Among the English **vessels** was a warship by the name of Mary Rose. Built in Portsmouth some 35 years earlier, she had had a long and successful fighting career, and was a favourite of King Henry VIII. Accounts of what happened to the ship **vary**: while **witnesses** agree that she was not hit by the French, some **maintain** that she

**Fleet**= ship in a navy flit

**Engaged in**= to be doing or to become involved in an activity in'geɪdʒd ɪn

**Vessel**= a ship or large boat 'vesəl

**Vary**= differ, be different 'veəri

**Witness**= observe 'wɪtnəs

**Maintain**= remain, make something continue at the same level, standard, etc. meɪn'teɪn

was **outdated**, **overladen** and sailing too low in the water, others that she was **mishandled** by **undisciplined crew**. What is **undisputed**, however, is that the Mary Rose sank into the Solent that day, taking at least 500 men with her. After the battle, attempts were made to recover the ship, but these failed.

The Mary Rose came to rest on the seabed, lying on her starboard (right) side at an angle of approximately 60 degrees. The hull (the body of the ship) **acted as a** trap for the sand and mud carried by Solent **currents**. As a result, the starboard side filled rapidly, leaving the **exposed** port (left) side to be **eroded** by **marine organisms** and **mechanical degradation**. Because of the way the ship sank, nearly all of the starboard half survived **intact**. During the seventeenth and eighteenth centuries, the entire site became covered with a layer of hard grey clay, which **minimised** further **erosion**.

Then, on 16 June 1836, some fishermen in the Solent found that their equipment was caught on an underwater **obstruction**, which turned out to be the Mary Rose. Diver John Deane happened to be exploring another **sunken** ship nearby, and the fishermen approached him, asking him to free their **gear**. Deane **dived** down, and found the equipment caught on a **timber protruding** slightly from the seabed. Exploring further, he **uncovered** several other timbers and a bronze gun. Deane continued diving on the site **intermittently** until 1840, recovering several more guns, two bows, various timbers, part of a pump and various other small finds.

The Mary Rose then **faded into obscurity** for another hundred years. But in 1965, **military** historian and

**Outdated**=old-fashioned, old, obsolete  
# current, modern, fashionable *aut'deɪtɪd*  
**Overladen**= filled with too many people or things *əʊvə'leɪdn*  
**Mishandle**= to treat something roughly, often causing damage *mɪs'hændl*  
**Undisciplined** = behaving in an uncontrolled way # disciplined, well-behaved *ʌn'dɪsɪplɪnd*  
**Crew**= sailors, seamen *kru:*  
**Undisputed**= acknowledged, undeniable *ʌndɪ'spju:tɪd*

**Act as sth**= do a particular job *ækt əz 'sʌmθɪŋ*  
**Current** = a movement of water in a particular direction *'kʌrənt*  
**Exposed** = not covered, unprotected, *ɪk'spəʊzd*  
**To be eroded** = to be weakened= to be damaged *tu bi i'ruədɪd*  
**Marine** = sea, oceanic, aquatic *mə'ri:n*  
**Organism**= living thing, creature *'ɔ:ɡənɪzəm*  
**Mechanical** = connected with machines and engines *mɪ'kænɪkəl*  
**Degradation**= the process by which something changes to a worse condition *deɡre'deɪʃən*  
**Intact**= unbroken, unharmed, undamaged  
# broken, damaged *ɪn'tækt*  
**Minimize**= reduce, decrease, cut down, lessen  
# maximize, increase *'mɪnɪmaɪz*

**Obstruction**= blockage, obstacle *əb'strʌkʃən*  
**Sunken** = having fallen to the bottom of the sea *'sʌŋkən*  
**Gear**= a set of equipment or tools you need for a particular activity *ɡɪə*  
**Dive**= swim under water, go under water *daɪv*  
**Timber**= wood *'tɪmbə*  
**Protrude**= stick out from or through sth *prə'tru:d*  
**Uncover**= detect, discover, find out  
# cover, conceal, hide *ʌn'kʌvə*  
**Intermittently** = stopping and starting repeatedly or with periods in between *ɪntə'mɪtnli*

**Fade into obscurity**= being in the state of not being known or remembered *feɪd 'ɪntə əb'skjʊərɪti*  
**Military** = used by, involving, or relating to the army, navy, or air force *'mɪləteri*

**amateur** diver Alexander McKee, in **conjunction** with the British Sub-Aqua Club, **initiated** a project called 'Solent Ships'. While on paper this was a plan to examine a number of known **wrecks** in the Solent, what McKee really hoped for was to find the Mary Rose. **Ordinary** search techniques proved **unsatisfactory**, so McKee entered into **collaboration** with Harold E. Edgerton, professor of electrical engineering at the Massachusetts Institute of Technology. In 1967, Edgerton's side-scan **sonar** systems **revealed** a large, unusually shaped object, which McKee believed was the Mary Rose.

Further **excavations** revealed stray pieces of timber and an iron gun. But the **climax** to the operation came when, on 5 May 1971, part of the ship's **frame** was uncovered. McKee and his team now knew for certain that they had found the **wreck**, but were as yet unaware that it also **housed** a **treasure trove** of beautifully preserved **artefacts**. Interest in the project grew, and in 1979, The Mary Rose Trust was formed, with Prince Charles as its President and Dr Margaret Rule its Archaeological Director. The decision whether or not to **salvage** the wreck was not an easy one, although an excavation in 1978 had shown that it might be possible to raise the hull. While the original aim was to raise the hull if at all **feasible**, the operation was not given the **go-ahead** until January 1982, when all the necessary information was available. An important factor in trying to **salvage** the Mary Rose was that the remaining hull was an open shell. This led to an important decision being taken: namely to carry out the lifting operation in three very **distinct** stages. The hull was **attached** to a lifting frame via a network of bolts and lifting wires. The problem of the hull being sucked back downwards into the mud was **overcome** by using 12 hydraulic **jacks**. These raised it a few centimetres over a period of several days, as the lifting frame rose slowly up its four legs. It was only when the hull was hanging freely from the lifting frame, clear of

**Amateur** # professional 'amətə  
**Conjunction**=combination kən'dʒʌŋkʃən  
**Initiate**= start, set off i'nɪʃieɪt  
**Wreck**= a ship that has sunk rek  
**Ordinary**= normal, usual, regular, common  
'ɔːdəneri  
**Unsatisfactory** = not good enough or not acceptable ʌnsætəs'fæktəri  
**Collaboration**= teamwork, partnership kələbə'reɪʃən  
**Sonar**= relating to sun 'səʊnə:  
**Reveal**= discover, make known  
# hide, conceal ri'vi:l

**Excavation**= digging ekskə'veɪʃn  
**Climax**= peak, highest point 'klaɪmæks  
**Frame**= structure freɪm  
**Wreck** = a ship that has sunk rek  
**House** = provide space for something haʊs  
**Treasure trove**= a group of valuable or interesting things 'treʒə trəʊv  
**Artefacts**= historical objects 'ɑːtɪfæks

**Salvage** = to save goods from damage or destruction, especially from a ship that has sunk 'sælvɪdʒ

**Feasible**= possible, practicable, workable... # impractical 'fiːzəbəl

**Go-ahead** = an occasion when permission is given for someone to start doing something 'gəʊ əhed

**Distinct** = clearly different or belonging to a different type dɪ'stɪŋkt

**Attach**= glue, join, connect ə'tætʃ

**Overcome** = defeat or succeed in controlling or dealing with something əʊvə'kʌm

**Jack** = a piece of equipment used to lift a heavy weight off the ground, such as a car, and support it while it is in the air dʒæk

the seabed and the suction effect of the surrounding mud, that the salvage operation progressed to the second stage. In this stage, the lifting frame was fixed to a **hook** attached to a **crane**, and the hull was lifted completely clear of the seabed and transferred underwater into the lifting cradle. This required **precise** positioning to locate the legs into the '**stabbing guides**' of the lifting **cradle**. The lifting cradle was designed to fit the hull using archaeological survey drawings, and was fitted with air bags to provide additional cushioning for the hull's delicate timber **framework**. The third and final stage was to lift the entire structure into the air, by which time the hull was also supported from below. Finally, on 11 October 1982, millions of people around the world **held their breath** as the timber **skeleton** of the Mary Rose was lifted clear of the water, ready to be returned home to Portsmouth.

**Hook**= a curved piece of metal or plastic that you use for hanging things on *hʊk*  
**Crane**= a large tall machine used by builders for lifting heavy things *kreɪn*  
**Precise**= exact, specific, accurate  
 # imprecise, inaccurate *prɪ'saɪs*  
**Stabbing** = very sharp, sudden and strong  
*'stæbɪŋ*  
**Guide**= instruction= direction *gaɪd*  
**Cradle** = a structure that is used to lift something heavy up or down *'kreɪdl*  
**Framework**= the main supporting parts of a building, vehicle, or object *'freɪmwɜ:k*  
**Hold (one's) breath (idiom)**= wait for something specific to happen *'həʊld (wʌnz) breθ ('ɪdɪəm)*  
**Skeleton**= the main structure that supports a building, bridge etc. *'skelətən*

## READING PASSAGE 2



A.

**E**aster Island, or Rapu Nui as it is known locally, is home to several hundred **ancient** human statues - the moai. After this **remote** Pacific island was **settled** by the Polynesians, it remained **isolated** for centuries. All the energy and resources that went into the moai - some of which are ten metres tall and weigh over 7,000 kilos - came from the island itself. Yet when Dutch explorers landed in 1722, they met a Stone Age culture. The moai were **carved** with stone tools, then transported for many kilometres, without the use of animals or wheels, to **massive** stone **platforms**. The **identity** of the moai builders was in doubt until well into the twentieth century. Thor Heyerdahl, the Norwegian ethnographer and adventurer, thought the statues had been created by pre-Inca peoples from Peru. Bestselling Swiss author Erich von Daniken believed they were built by **stranded extraterrestrials**. Modern science - linguistic, **archaeological** and **genetic**

**Civilisation**= human development  
(*civ=citizen .i.e, civic, civil*) sɪvəl-aɪ'zeɪʃən  
**Ancient**= very old 'eɪnʃənt  
**Remote**= distant, far-off # central ɪ'məʊt  
**Settle**= set up home, emigrate to 'setl  
**Isolated**= remote 'aɪsələɪtɪd  
**Carve**= to make something by cutting into especially wood or stone, or to cut into the surface of stone, wood, etc.kɑ:v  
**Massive**= very large, solid, and heavy 'mæsɪv  
**Platform**= stage 'plætfɔ:m  
**Identity**= who a person is aɪ'dentəti  
**Stranded**= stuck 'strændɪd  
**Extraterrestrial**= a creature that people think may exist on another planet (*terr=earth.i.e, territoty, terrain*) ekstrə'terɪstriəl  
**Archaeological**= belonging to the study of ancient societies by examining what remains of their buildings, graves, tools etc. ɑ:kɪə'lɒdʒɪkəl  
**Genetic**= relating to genes or genetics dʒə'netɪk

evidence - has **definitively** proved the moai builders were Polynesians, but not how they moved their creations. Local **folklore** maintains that the statues walked, while researchers have tended to **assume** the ancestors **dragged** the statues somehow, using **ropes** and **logs**.

B.  
When the Europeans arrived, Rapa Nui was grassland, with only a few scrawny trees. In the 1970s and 1980s, though, researchers found **pollen** preserved in lake **sediments**, which proved the island had been covered in lush palm forests for thousands of years. Only after the Polynesians arrived did those forests disappear. US scientist Jared Diamond believes that the Rapanui people - **descendants** of Polynesian settlers - **wrecked** their own environment. They had unfortunately settled on an extremely **fragile** island - dry, cool, and too remote to be properly **fertilised** by windblown volcanic **ash**. When the islanders cleared the forests for firewood and farming, the forests didn't grow back. As trees became **scarce** and they could no longer construct wooden canoes for fishing, they ate birds. Soil **erosion** decreased their crop yields. Before Europeans arrived, the Rapanui had **descended** into civil war and cannibalism, he maintains. The **collapse** of their isolated civilisation, Diamond writes, is a 'worst-case **scenario** for what may lie ahead of us in our own future'.

C.  
The moai, he thinks, **accelerated** the self-**destruction**. Diamond interprets them as power **displays** by **rival chieftains** who, trapped on a remote little island, lacked other ways of **asserting** their **dominance**. They competed by building ever bigger figures. Diamond thinks they laid the moai on wooden sledges, **hauled** over log rails, but that required both a lot of wood and a lot of people. To **feed** the people, even more land had to be cleared. When the wood was gone and civil war

**Definitively**=perfectly *dɪˈfɪnɪtɪvli*  
**Folklore**= myths, legends *ˈfəʊklɔː*  
**Assume**= presume, to think or accept that something is true without having proof *əˈsjʊːm*  
**Drag**= pull *dræɡ*  
**Rope**= very strong thick string, made by twisting together many thinner strings *rəʊp*  
**Log**= a thick piece of wood from a tree *lɒɡ*

**Pollen**= a fine powder produced by flowers, which is carried by the wind or by insects to other flowers of the same type, making them produce seeds *ˈpɒlən*  
**Sediment**= solid substances that settle at the bottom of a liquid *ˈsedəmənt*  
**Descendant**= offspring, previous generation *dɪˈsendənt*  
**Wreck**= ruin, destroy, damage *rek*  
**Fragile**= easily broken # strong *ˈfrædʒaɪl*  
**Fertilise**= to spread a natural or chemical substance on land or plants, in order to make the plants grow well *ˈfɜːtəlaɪz*  
**Ash**= the soft grey powder that remains after something has been burned *æʃ*  
**Scarce**= rare, insufficient # plentiful, abundant *skeəs*  
**Erosion**= the fact of soil, stone, etc. being gradually damaged and removed by the waves, rain, or wind *ɪˈrəʊʒən*  
**Descend**= fall # ascend *dɪˈsend*  
**Collapse**= breakdown, failure, end *kəˈlæps*  
**Scenario**= situation *səˈnɑːriəʊ*

**Accelerate**= quicken, speed up # slow down *əkˈseləreɪt*  
**Destruction**= the act or process of destroying something *dɪˈstrʌkʃən*  
**Display**= an arrangement of things *dɪˈspleɪ*  
**Rival**= opponent # ally *ˈraɪvəl*  
**Chieftain**= the leader of a tribe *ˈtʃiːftɪn*  
**Assert**= to make other people recognize your right or authority *əˈsɜːt*  
**Dominance**= superiority, power, authority *ˈdɒmənəns*  
**Haul**= drag, pull, tug *hɔːl*  
**Feed**= give food to, provide food for *fiːd*

began, the islanders began **toppling** the moai. By the nineteenth century none were standing.

D.

Archaeologists Terry Hunt of the University of Hawaii and Carl Lipo of California State University agree that Easter Island lost its lush forests and that it was an 'ecological **catastrophe**' - but they believe the islanders themselves weren't to blame. And the moai certainly weren't. Archaeological excavations indicate that the Rapanui went to heroic efforts to protect the resources of their wind-lashed, **infertile** fields. They built thousands of circular stone **windbreaks** and gardened inside them, and used broken volcanic rocks to keep the soil **moist**. In short, Hunt and Lipo argue, the prehistoric Rapanui were **pioneers** of **sustainable** farming.

E.

Hunt and Lipo **contend** that moai-building was an activity that helped keep the peace between islanders. They also believe that moving the moai required few people and no wood, because they were walked **upright**. On that issue, Hunt and Lipo say, archaeological evidence backs up Rapanui folklore. Recent experiments indicate that as few as 18 people could, with three strong ropes and a bit of practice, easily **manoeuvre** a 1,000 kg moai **replica** a few hundred metres. The figures' fat bellies **tilted** them forward, and a D-shaped base allowed handlers to roll and rock them side to side.

F.

Moreover, Hunt and Lipo are **convinced** that the settlers were not **wholly** responsible for the loss of the island's trees. Archaeological finds of **nuts** from the **extinct** Easter Island palm show tiny **grooves**, made by the teeth of Polynesian rats. The rats arrived along with the settlers, and in just a few years, Hunt and Lipo calculate, they would have **overrun** the island. They would have prevented the reseedling of the slow-growing palm trees and thereby **doomed** Rapa Nui's forest, even without the settlers' campaign of **deforestation**. No doubt the rats ate birds' eggs too. Hunt and Lipo also see no evidence that Rapanui

**Topple**= to take power away from a leader or government, especially by force 'tɒpəl

**Catastrophe**= disaster kə'tæstrəfi

**Infertile**= unproductive # fertile ɪn'fɜ:təl

**Windbreak** = something that gives protection from the wind, such as a row of trees, bushes, or a wall 'wɪndbreɪk

**Moist**= wet, damp #dry mɔɪst

**Pioneer**= leader paɪə'nɪə

**Sustainable**= able to continue for a long time sə'steɪnəbəl

**Contend**= argue, claim, assert kən'tend

**Upright**= straight up 'ʌpraɪt

**Manoeuvre**= move, turn mə'nu:və

**Replica**= copy, duplicate, reproduction. 'replɪkə

**Tilt**= move, tip tɪlt

**Convince**= persuade kən'vɪns

**Wholly** = completely 'həʊl-li

**Extinct**= no longer in existence ɪk'stɪŋkt

**Nut**= seed nʌt

**Groove**= a thin line cut into a hard surface. gru:v

**Overrun**= if a place is overrun by unwanted things or people, they spread over it in great numbers. əʊvə'rʌn

**Doom**= ruin, destroy du:m

**Deforestation**= the cutting or burning down of all the trees in an area di:fore'steɪʃən

civilisation **collapsed** when the palm forest did. They think its population grew rapidly and then remained more or less **stable** until the arrival of the Europeans, who introduced deadly diseases to which islanders had no **immunity**. Then in the nineteenth century **slave** traders **decimated** the population, which **shrivelled** to 111 people by 1877.

G.

Hunt and Lipo's vision, therefore, is one of an island populated by peaceful and **ingenious** moai builders and careful **stewards** of the land, rather than by **reckless** destroyers ruining their own environment and society. 'Rather than a case of **abject failure**, Rapa Nui is an unlikely story of success', they claim. Whichever is the case, there are surely some valuable lessons which the world at large can learn from the story of Rapa Nui.

**Collapse**= fall down kə'ləps

**Stable**= fixed = steady #changeable 'steɪbəl

**Immunity**= resistance, protection ɪ'mju:nəti

**Slave**= someone who is owned by another person and works for them for no money.

slɛɪv

**Decimate**= destroy, devastate, ruin 'desɪmeɪt

**Shrivel** = become smaller, shrink 'ʃrɪvəl

**Ingenious**= clever, good at inventing ɪn'dʒɪniəs

**Steward**= manager, keeper, guardian, supervisor 'stju:əd

**Reckless**= careless # careful, cautious 'rekləs

**Abject failure**= the state of being extremely poor, unhappy, unsuccessful etc. 'æbdʒekt 'feɪljə

## READING PASSAGE 3:



**A**n emerging discipline called neuroaesthetics is seeking to bring scientific **objectivity** to the study of art, and has already given us a better understanding of many **masterpieces**. The **blurred** imagery of Impressionist paintings seems to stimulate the brain's **amygdala**, for instance. Since the amygdala plays a crucial role in our feelings, that finding might explain why many people find these pieces so **moving**.

Could the same approach also shed light on **abstract** twentieth-century pieces, from Mondrian's geometrical blocks of colour, to Pollock's seemingly haphazard arrangements of splashed paint on canvas? **Sceptics** believe that people claim to like such works simply

**Neuroaesthetics**= a field of study in which researchers attempt to understand how the brain responds to art (*neur(o)*= relating to nerves, i.e: *neuroscience*) 'njuərəʊ.i:s, θetɪks

**Emerging**= rising ɪ'mɜ:dʒɪŋ

**Discipline**= field of study, branch of knowledge, subject, area 'dɪsɪplɪn

**Objectivity**= fairness, open-mindedness

# subjectivity ɒbdʒek'tɪvəti

**Masterpiece**= a work of art, a piece of writing or music etc that is of very high quality or that is the best of a particular artist, writer. 'mɑ:stəpi:s

**Blurred**= unclear # clear, distinct blɜ:d

**Stimulate**= quicken, accelerate, arouse, inspire 'stɪmjəleɪt

**Amygdala**= one of two parts of the brain that affect how people feel emotions, especially fear and pleasure ə'mɪgdələ

**Moving**= touching, affecting, emotional 'mu:vɪŋ

**Shed light on**= make clear, explain, simplify ʃed laɪt ɒn

**Abstract**= non-realistic, symbolic 'æbstrækt

**Haphazard**= happening or done in a way that is not planned or organized hæp'hæzəd

**Sceptic**= a person who disagrees with particular claims and statements 'skeptɪk

because they are famous. We certainly do have an **inclination** to follow the crowd. When asked to make simple **perceptual** decisions such as matching a shape to its rotated image, for example, people often choose a definitively wrong answer if they see others doing the same. It is easy to imagine that this **mentality** would have even more impact on a **fuzzy** concept like art **appreciation**, where there is no right or wrong answer.

Angelina Hawley-Dolan, of Boston College, Massachusetts, responded to this **debate** by asking volunteers to view pairs of paintings - either the creations of famous abstract artists or the **doodles** of infants, chimps and elephants. They then had to judge which they preferred. A third of the paintings were given no **captions**, while many were labelled incorrectly - volunteers might think they were viewing a chimp's **messy brushstrokes** when they were actually seeing an **acclaimed** masterpiece. In each set of trials, volunteers generally preferred the work of **renowned** artists, even when they believed it was by an animal or a child. It seems that the viewer can sense the artist's vision in paintings, even if they can't explain why.

Robert Pepperell, an artist based at Cardiff University, creates **ambiguous** works that are neither entirely abstract nor clearly representational. In one study, Pepperell and his **collaborators** asked volunteers to decide how 'powerful' they considered an artwork to be, and whether they saw anything familiar in the piece. The longer they took to answer these questions, the more highly they rated the piece under **scrutiny**, and the greater their neural activity. It would seem that the brain sees these images as puzzles, and the harder it is to **decipher** the meaning, the more rewarding is the moment of **recognition**.

And what about artists such as Mondrian, whose paintings consist **exclusively** of horizontal and vertical

**Inclination**= a feeling that makes you want to do something = tendency, proclivity  
ɪŋkləˈneɪʃən

**Perceptual**= relating to the ability to notice something or come to an opinion about something using your senses  
pəˈseptʃuəl

**Mentality**= a particular attitude or way of thinking  
menˈtæləti

**Fuzzy**= unclear  
ˈfʌzi

**Appreciation**= a feeling of being grateful for something someone has done  
əˈpriːʃiˈeɪʃən

**Debate**= discussion  
dɪˈbeɪt

**Doodle**= a rough drawing  
ˈduːdl

**Caption**= title, description  
ˈkæpʃən

**Messy** = disordered, chaotic, confused, disorganized  
ˈmesi

**Brushstroke** = the way in which something, especially paint, is put on to a surface with a brush  
ˈbrʌʃstroʊk

**Acclaimed**= honored, admired, praised  
əˈkleɪmd

**Renowned**= famous, well-known  
rɪˈnaʊnd

**Ambiguous**= uncertain, confusing, unclear  
æmˈbɪɡjuəs

**Collaborator**= coworkers, colleague, partner (*co*= together; with; i.e. *coexist*, *co-ownership*)  
kəˈlæbəreɪtə

**Scrutiny**= examination, analysis  
ˈskruːtɪni

**Decipher**= decode, interpret  
dɪˈsaɪfə

**Recognition**= the act of realizing and accepting that smt is true or important  
rekəɡˈnɪʃən

**Exclusively**= only  
ɪkˈskluːsɪv.li

lines encasing blocks of colour? Mondrian's works are **deceptively simple**, but eye-tracking studies confirm that they are **meticulously** composed, and that simply rotating a piece radically changes the way we view it. With the originals, volunteers' eyes tended to stay longer on certain places in the image, but with the **altered** versions they would **flit** across a piece more rapidly. As a result, the volunteers considered the altered versions less **pleasurable** when they later rated the work.

In a similar study, Oshin Vartanian of Toronto University asked volunteers to compare original paintings with ones which he had altered by moving objects around within the **frame**. He found that almost everyone preferred the original, whether it was a Van Gogh still life or an abstract by Miro. Vartanian also found that changing the **composition** of the paintings reduced activation in those brain areas linked with meaning and **interpretation**.

In another experiment, Alex Forsythe of the University of Liverpool analysed the visual **intricacy** of different pieces of art, and her results suggest that many artists use a key level of detail to **please** the brain. Too little and the work is boring, but too much results in a kind of 'perceptual overload', according to Forsythe. What's more, **appealing** pieces both abstract and representational, show signs of 'fractals' - repeated **motifs recurring** in different scales, fractals are common throughout nature, for example in the shapes of mountain peaks or the branches of trees. It is possible that our visual system, which evolved in the great outdoors, finds it easier to process such patterns.

It is also **intriguing** that the brain appears to process movement when we see a handwritten letter, as if we are replaying the writer's moment of creation. This has led some to **wonder** whether Pollock's works feel so **dynamic** because the brain **reconstructs** the

**Deceptively simple**= looking simple, but actually the opposite dɪ'sep.tɪv.li 'sɪmpəl

**Meticulously**= carefully # carelessly mə'tɪkjələsli

**Altered**= changed

# original 'ɔ:l.təd

**Flit**= to move lightly or quickly and not stay in one place for very long flɪt

**Pleasurable**= enjoyable 'pleʒərəbəl

**Frame**= a border that surrounds and supports a picture, door, or window freɪm

**Composition**= the way that people or things are arranged in a painting or photograph kəm.pə'zɪʃən

**Interpretation**= explanation, understanding ɪn'tɜ:pri'teɪʃən

**Intricacy**= complexity 'ɪntrɪkəsi

**Please** = satisfy pli:z

**Perceptual**= relating to the ability to notice something or come to an opinion about something using your senses pə'sep.tʃu.əl

**Appealing**= attractive, interesting, tempting, charming ə'pi:lɪŋ

**Motif**= pattern məʊ'ti:f

**Recur** = occur again, be repeated rɪ'kɜ:

**Intriguing**= fascinating, interesting, exciting ɪn'trɪ:ɡɪŋ

**Wonder**= doubt, question, self-ask 'wʌndə

**Dynamic**= lively, energetic daɪ'næmɪk

energetic actions the artist used as he painted. This may be down to our brain's 'mirror neurons', which are known to **mimic** others' actions. The **hypothesis** will need to be thoroughly tested, however. It might even be the case that we could use neuroaesthetic studies to understand the **longevity** of some pieces of artwork. While the fashions of the time might shape what is currently popular, works that are best adapted to our visual system may be the most likely to **linger** once the trends of previous generations have been forgotten.

It's still early days for the field of neuroaesthetics - and these studies are probably only a taste of what is to come. It would, however, be **foolish** to reduce art **appreciation** to a set of scientific laws. We shouldn't **underestimate** the importance of the style of a particular artist, their place in history and the artistic environment of their time. Abstract art offers both a challenge and the freedom to play with different interpretations. In some ways, it's not so different to science, where we are **constantly** looking for systems and **decoding** meaning so that we can view and appreciate the world in a new way.

**Reconstruct**= rebuild ri:kən'strʌkt

**Mimic**= copy, imitate 'mɪmɪk

**Hypothesis** = theory, assumption  
haɪ'pɒθəsɪs

**Longevity**= a long life lɒn'dʒevəti

**Linger**= remain # leave 'lɪŋgə

**Foolish**= stupid, silly, unwise 'fu:lɪʃ

**Appreciation**= admiration, enjoyment  
ə'pri:ʃi'eɪʃən

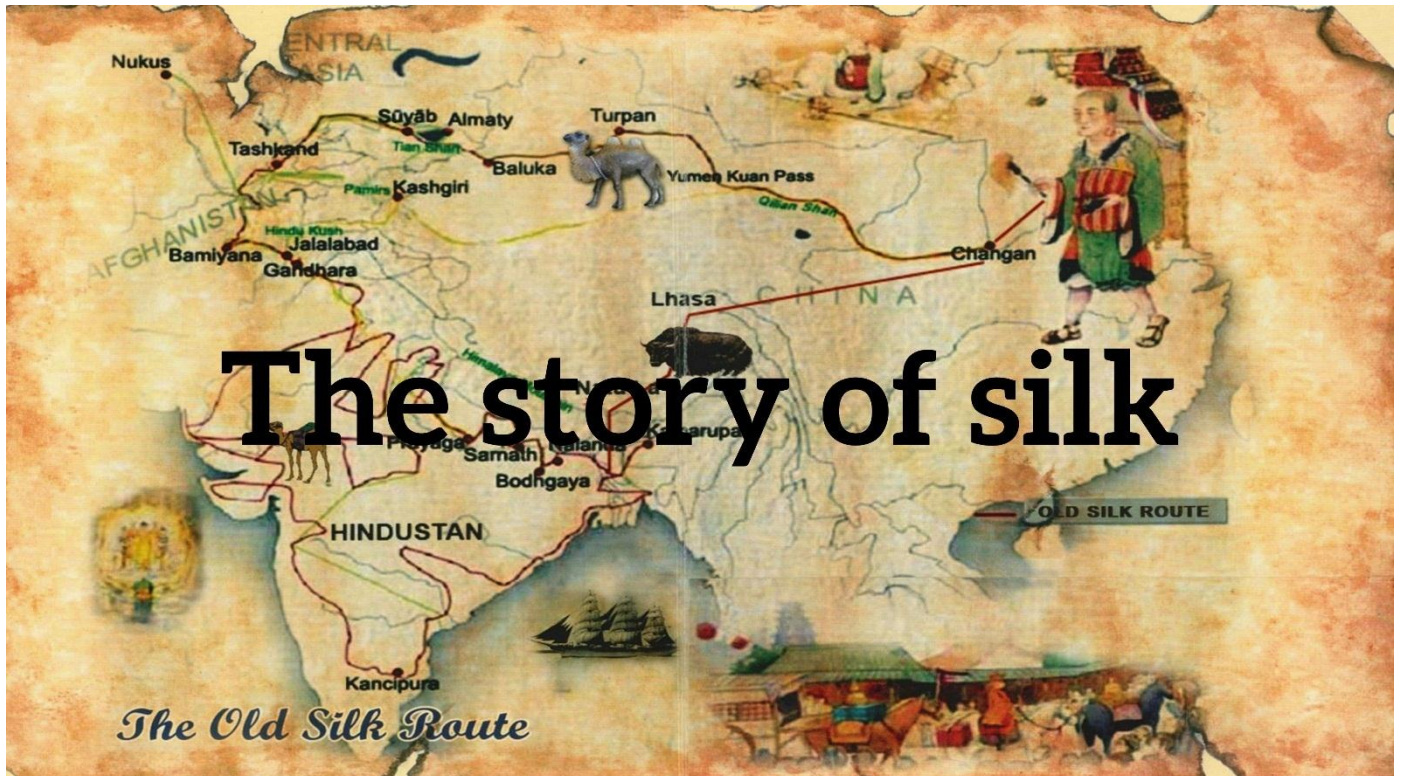
**Underestimate**= undervalue  
# overestimate (*under*= not enough, below, i.e.:  
*underperforming*) ʌndər'estɪmeɪt

**Constantly**= continually, all the time, or very often 'kɒn.stənt.li

**Decode**= decipher, interpret di:'kəʊd

## Test 3

### READING PASSAGE 1



**T**he history of the world's most **luxurious fabric**, from ancient China to the present day

Silk is a **fine, smooth** material produced from the cocoons - soft protective shells - that are made by mulberry silkworms (insect larvae). **Legend** has it that it was Lei Tzu, wife of the Yellow Emperor, ruler of China in about 3000 BC, who discovered silkworms. One account of the story goes that as she was taking a walk in her husband's gardens, she discovered that silkworms were **responsible** for the **destruction** of several mulberry trees. She collected a number of cocoons and sat down to have a rest. It just so happened that while she was sipping some tea, one of

**Luxurious**= expensive, costly /lʌg'zjʊəriəs

**Fabric**= material, cloth 'fæbrɪk

**Fine**= well, excellent, top quality faɪn

**Smooth**= downy, soft, flat smu:ð

**Legend**= myth, fairy tale 'ledʒənd

**Responsible**= being the primary cause of something (spon=guarantee, assurance.i.e sponsus, spondere) rɪ'spɒnsəbəl

**Destruction**= ruin, damage  
# construction dɪ'strʌkʃən

the cocoons that she had collected **landed in** the hot tea and started to **unravel** into a fine **thread**. Lei Tzu found that she could **wind** this thread around her fingers. Subsequently, she **persuaded** her husband to allow her to **rear** silkworms on a **grove** of mulberry trees. She also **devised** a special **reel** to draw the **fibres** from the cocoon into a single thread so that they would be strong enough to be woven into **fabric**. While it is unknown just how much of this is true, it is certainly known that silk **cultivation** has existed in China for several millennia.

Originally, silkworm farming was solely **restricted to** women, and it was they who were responsible for the growing, harvesting and weaving. Silk quickly grew into a symbol of status, and originally, only **royalty** were **entitled to** have clothes made of silk. The rules were gradually relaxed over the years until finally during the Qing Dynasty (1644—1911 AD), even **peasants**, the lowest **caste**, were also entitled to wear silk. Sometime during the Han Dynasty (206 BC-220 AD), silk was so **prized** that it was also used as a unit of **currency**. Government officials were paid their salary in silk, and farmers paid their taxes in grain and silk. Silk was also used as **diplomatic** gifts by the **emperor**. Fishing lines, bowstrings, musical instruments and paper were all made using silk. The earliest **indication** of silk paper being used was discovered in the **tomb** of a **noble** who is estimated to have died around 168 AD.

**Demand** for this **exotic** fabric eventually created the **lucrative** trade route now known as the Silk Road, taking silk westward and bringing gold, silver and wool to the East. It was named the Silk Road after its most **precious commodity**, which was considered to be worth more than gold. The Silk Road stretched over 6,000 kilometres from Eastern China to the Mediterranean Sea, following the Great Wall of China, climbing the Pamir mountain range, crossing modern-

**Land in**= fall into lænd ɪn  
**Unravel**= untie, loosen, separate out ʌnˈrævəl  
**Thread**= a long thin string of cotton, silk etc used to sew or weave cloth θred  
**Wind**= roll, twist wɪnd  
**Persuade**= convince pəˈsweɪd  
**Rear**= raise, nurture rɪər  
**Grove**= a group of trees planted close together grəʊv  
**Devise**= invent, come up with, formulate, design dɪˈvaɪz  
**Reel**= a round object around which you wind such things as thread ri:l  
**Fibre**= a mass of threads used to make rope, cloth etc. ˈfaɪbə  
**Fabric**= cloth or material for making clothes, covering furniture ˈfæbrɪk  
**Cultivation**= growing, planting kʌltəˈveɪʃən

**Restrict something to something**= limit rɪˈstrɪkt ˈsʌmθɪŋ tə ˈsʌmθɪŋ  
**Royalty**= members of a royal family ˈrɔɪəlti  
**Be entitled to (do) something**= to have the right/ to be allowed to do smt bi ɪnˈtaɪ.təld tu: (du:) ˈsʌmθɪŋ  
**Peasant**= poor farmer ˈpezənt  
**Caste**= social class kɑ:st  
**Prized**= treasured, precious, valued praɪzd  
**Currency**= money ˈkʌrənsi  
**Diplomatic**= relating to the work of diplomats who officially represent their government in a foreign country dɪpləˈmætɪk  
**Emperor**= royal leader, monarch ˈempərə  
**Indication**= sign, mark ɪndɪˈkeɪʃən  
**Tomb**= burial place for last resting place tu:m  
**Noble**= a member of the highest social class ˈnəʊbəl

**Exotic**= unusual, unconventional, extravagant, unfamiliar ɪgˈzɒtɪk

**Lucrative**= profitable, thriving, flourishing ˈlu:kreɪtɪv

**Precious**= valuable, important ˈpreʃəs

**Commodity**= product, item kəˈmɒdəti

day Afghanistan and going on to the Middle East, with a major trading market in Damascus. From there, the **merchandise** was shipped across the Mediterranean Sea. Few **merchants** travelled the entire route; goods were handled mostly by a series of **middlemen**.

With the mulberry silkworm being native to China, the country was the world's sole producer of silk for many hundreds of years. The secret of silk-making eventually reached the rest of the world via the Byzantine Empire, which ruled over the Mediterranean **region** of southern Europe, North Africa and the Middle East during the period 330—1453 AD. According to another legend, monks working for the Byzantine emperor Justinian **smuggle** silkworm eggs to Constantinople (Istanbul in modern-day Turkey) in 550 AD, **concealed** inside **hollow** bamboo walking canes. The Byzantines were as **secretive** as the Chinese, however, and for many centuries the weaving and trading of silk fabric was a strict **imperial monopoly**. Then in the seventh century, the Arabs **conquered** Persia, **capturing** their **magnificent** silks in the process. Silk production thus spread through Africa, Sicily and Spain as the Arabs swept, through these lands. Andalusia in southern Spain was Europe's main silk-producing centre in the tenth century. By the thirteenth century, however, Italy had become Europe's leader in silk production and **export**. Venetian merchants traded extensively in silk and encouraged silk growers to **settle in** Italy. Even now, silk processed in the province of Como in northern Italy enjoys an **esteemed reputation**.

The nineteenth century and industrialisation saw the downfall of the European silk industry. Cheaper Japanese silk, trade in which was greatly **facilitated** by the opening of the Suez Canal, was one of the many factors driving the trend. Then in the twentieth century, new **manmade** fibres, such as nylon, started to be used in what had traditionally been silk products, such as stockings and parachutes. The two world wars,

**Merchandise**= goods, products 'mɜːtʃəndaɪz  
**Merchant**= businessman, tradesman 'mɜːtʃənt  
**Middlemen**= someone who buys things in order to sell them to someone else, or who helps to arrange business deals for other people 'mɪdlmæn

**Region**=an area of a country, especially one that has a particular characteristic or is known for something 'rɪdʒən

**Smuggle**= take/send/bring goods/people secretly and illegally into/out of a country 'smʌɡəl

**Conceal**= hide, cover kən'siːl

**Hollow**= having a hole or empty space inside 'hɒləʊ

**Secretive**= a secretive person or organization likes to keep their thoughts, intentions, or actions hidden from others 'siːkrətɪv

**Imperial**= relating to an empire or to the person who rules it ɪm'piəriəl

**Monopoly**= domination= exclusiveness (mono-=only.i.e **monotone**, **monist**) mə'nɒpəli

**Conquer**= defeat, beat, overpower 'kɒŋkə

**Capture**= take over, take 'kæptʃə

**Magnificent**= great, fine, wonderful (magn=big, large .i.e **magnitude**, **magnificien**) məg'nɪfəsənt

**Export**= sell abroad, sell overseas, sell to other countries # import 'eksɒɹt

**Settle in somewhere** = to go to live in a new place, and stay there for a long time 'setl ɪn 'sʌmwɛə

**Esteemed**= respected, admired, honored ɪ'stiːmd

**Reputation**= fame repjə'teɪʃən

**Facilitate**= make easy, make possible, enable fe'sɪlɪteɪt

**Manmade**= artificial, synthetic, manufactured # natural mæn'meɪd

which **interrupted** the supply of **raw** material from Japan, also **stifled** the European silk industry. After the Second World War, Japan's silk production was restored, with improved production and quality of raw silk. Japan was to remain the world's biggest producer of raw silk, and practically the only major exporter of raw silk, until the 1970s. However, in more recent decades, China has gradually **recaptured** its position as the world's biggest producer and exporter of raw silk and **silk yarn**. Today, around 125,000 metric tons of silk are produced in the world, and almost two thirds of that production takes place in China.

**Interrupt**= stop, end, halt ɪntəˈrʌpt  
**Raw**= unprocessed, unrefined, natural roː  
**Stifle**= prevent st from happening 'staɪfəl  
  
**Recapture**= bring back, take over again riːˈkæptʃə  
  
**Silk yarn**= thread used for making cloth or for knitting sɪlk jɑːn

## READING PASSAGE 2



**A**nimal migration, however it is **defined**, is far more than just the movement of animals. It can loosely be described as travel that takes place at regular intervals - often in an annual cycle - that may involve many members of a species, and is rewarded only after a long journey. It suggests **inherited instinct**. The biologist Hugh Dingle has **identified** five characteristics that apply, in varying degrees and combinations, to all migrations. They are **prolonged** movements that carry animals outside familiar habitats; they tend to be linear, not zigzaggy; they involve special behaviours concerning preparation (such as **overfeeding**) and arrival; they demand special **allocations** of energy. And one more: migrating animals maintain an intense attentiveness to the greater mission, which keeps them **undistracted** by

**Migration**= relocation, resettlement, movement *maɪ'greɪʃən*

**Define**= describe correctly and thoroughly *dɪ'faɪn*

**Inherited**= inborn, innate = having qualities, physical features that controlled by gene passed from their parent *ɪn'herɪtɪd*

**Instinct (n)**= nature, character *'ɪnstɪŋkt*

**Instinctive (adj)**= intuitive, natural, innate, inborn *ɪn'stɪŋktɪv*

**Identify**= discover, find, determine *aɪ'dentɪfaɪ*

**Prolonged**= expanded, lengthened *prə'lɒŋd*

**Linear**= following a straight line *'lɪniə*

**Overfeeding**= eating more than they need for immediate purposes (**over-** = *too much*; i.e: **oversensitive**, **overreact**) *əʊvə'fi:dɪŋ*

**Allocation**= distribution *ælə'keɪʃən*

**Undistracted** = able to concentrate fully on something *ʌndɪ'straktɪd*

**temptations** and **undeterred** by challenges that would turn other animals aside.

An arctic tern, on its 20,000 km flight from the extreme south of South America to the Arctic circle, will take no notice of a nice smelly herring offered from a bird-watcher's boat along the way. While local gulls will **dive voraciously** for such **handouts**, the tern flies on. Why? The arctic tern **resists distraction** because it is driven at that moment by an **instinctive** sense of something we humans find admirable: larger purpose. In other words, it is determined to reach its destination. The bird senses that it can eat, rest and **mate** later. Right now it is totally focused on the journey; its undivided **intent** is arrival. Reaching some gravelly coastline in the Arctic, upon which other arctic terns have **converged**, will serve its larger purpose as shaped by **evolution**: finding a place, a time, and a set of circumstances in which it can successfully **hatch** and **rear offspring**.

But migration is a **complex** issue, and biologists define it differently, depending in part on what sorts of animals they study. Joe! Berger, of the University of Montana, who works on the American pronghorn and other large **terrestrial mammals**, prefers what he calls a simple, practical definition suited to his **beasts**: 'movements from a seasonal home area away to another home area and back again'. Generally the reason for such seasonal back-and-forth movement is to **seek** resources that aren't available within a single area year-round.

But daily **vertical** movements by zooplankton in the ocean - upward by night to seek food, downward by day to escape **predators** - can also be considered **migration**. So can the movement of aphids when, having **depleted** the young leaves on one food plant,

**Temptation**= attraction, lure, appeal  
temp'teɪʃən

**Undeterred**= if you are undeterred by something, you do not allow it to stop you doing what you want ʌndɪ'tɜ:d

**Dive**= go underwater daɪv

**Voracious**= very eager for something, especially a lot of food və'reɪʃəs

**Handout**= something such as food, clothing, or money that is given free to someone who has a great need for it 'hændaʊt

**Resist**= avoid, refuse to accept, defend against rɪ'zɪst

**Distraction**= something that stops you paying attention to what you are doing dɪ'strækʃən

**Instinctive**= based on natural tendency, not involving thought ɪn'stɪŋktɪv

**Mate**= (of two animals or birds) to have sex in order to produce young meɪt

**Intent**= intention, aim, goal, target, purpose ɪn'tent

**Converge**= meet, come together kən'vɜ:dʒ

**Evolution**= growth, development ɪ:və'lʊ:ʃən

**Hatch**= (of a young bird, fish, insect, etc.) to come out of an egg hæʃ

**Rear**= look after, raise rɪə

**Offspring**= children 'ɒf,spɪŋ

**Complex**= complicated, difficult to understand 'kɒmpleks

**Terrestrial mammals**= animals giving birth to live young, not eggs and living on land rather than in the water or air tə'restriəl 'mæməlz

**Beast**= creature, animal bi:st

**Seek**= search for, look for si:k

**Vertical**= straight up= pointing up in a line that forms an angle of 90° with a flat surface # horizontal 'vɜ:tkəl

**Predator**= an animal that kills and eats other animals (prey). 'predətə

**Migration**= movement from one region to another and often back again, esp. according to the season of the year maɪ'greɪʃən

**Deplete**= reduce, eat up, lessen, exhaust, diminish # increase (de= down, reduce, i.e: destroy, degradation) dɪ'pli:t

their **offspring** then fly onward to a different host plant, with no one aphid ever returning to where it started. Dingle is an evolutionary biologist who studies insects. His definition is more **intricate** than Berger's, **citing** those five features that **distinguish** migration from other forms of movement. They allow for the fact that, for example, aphids will become **sensitive** to blue light (from the sky) when it's time for takeoff on their big journey, and sensitive to yellow light (reflected from **tender** young leaves) when it's **appropriate** to land. Birds will **fatten** themselves **with** heavy feeding in **advance** of a long migrational flight. The value of his definition, Dingle argues, is that it focuses attention on what the **phenomenon** of wildebeest migration shares with the phenomenon of the aphids, and therefore helps guide researchers towards understanding how **evolution** has produced them all.

Human behaviour, however, is having a **detrimental** impact on animal migration. The pronghorn, which **resembles** an antelope, though they are unrelated, is the fastest land mammal of the New World. One population, which spends the summer in the mountainous Grand Teton National Park of the western USA, follows a narrow route from its summer range in the mountains, across a river, and down onto the plains. Here they wait out the frozen months, feeding mainly on sagebrush blown clear of snow. These pronghorn are **notable** for the **invariance** of their migration route and the **severity** of its **constriction** at three **bottlenecks**. If they can't pass through each of the three during their spring migration, they can't reach their bounty of summer grazing; if they can't pass through again in autumn, escaping south onto those windblown **plains**, they are likely to die trying to overwinter in the deep snow. Pronghorn, dependent on distance vision and speed to keep safe from predators, **traverse** high, open shoulders of land, where they can see and run. At one of the bottlenecks, forested hills rise to form a V, leaving a **corridor** of open ground

**Offspring**= the young of animal 'ɒf, sprɪŋ  
**Intricate** = complicated, complex # simple 'ɪntrɪkət  
**Cite**= name, mention, refer to saɪt  
**Distinguish**= differentiate, discriminate dɪ'stɪŋɡwɪʃ  
**Sensitive to**= responsive to, reactive to, easily influenced by 'sensətɪv tə  
**Tender**= easily hurt or damaged 'tendə  
**Appropriate**= suitable, proper, fitting ə'prəʊpri-ət  
**Fatten sb with**= to become fatter by doing something. 'fætn 'sʌmbədi wɪð  
**In advance (of something)** = before something happens ɪn əd'vɑːns (əv 'sʌmθɪŋ)  
**Phenomenon**= a fact or situation that is observed to exist or happen frɪ'nɒmənən  
**Evolution**= the gradual change and development of an idea, situation, or object ɪ'veʊ'lʊːʃən  
  
**Detrimental**= harmful, negative, damaging # beneficial detre'mentl  
  
**Resemble**= look like, be similar to # differ rɪ'zembəl  
  
**be notable for**= be famous for bi 'nəʊtəbəl fə  
**Invariance**= constancy; stability # variance, difference ɪn'veəriəns  
**Severity**= dangerousness, extremity, harshness sə'verətɪ  
**Constriction**= restriction, limitation kən'strɪk.ʃən  
**Bottleneck**= a place in a road where the traffic cannot pass easily, so that there are a lot of delays 'bɒtlnek  
**Plain**= a large area of flat dry land pleɪn  
  
**Traverse**= cross, pass through, go over 'trævɜːs  
  
**Corridor**= long, narrow passage, passage way 'kɒrədɔː

only about 150 metres wide, filled with private homes. Increasing development is leading toward a **crisis** for the pronghorn, **threatening** to **choke off** their passageway.

**Conservation** scientists, along with some biologists and land managers within the USA's National Park Service and other **agencies**, are now working to **preserve** migrational behaviours, not just species and habitats. A National Forest has **recognised** the **path** of the pronghorn, much of which passes across its land, as a protected migration corridor. But neither the Forest Service nor the Park Service can control what happens on private land at a bottleneck. And with certain other migrating species, the challenge is complicated further - by vastly greater distances traversed, more **jurisdictions**, more borders, more dangers along the way. We will require **wisdom** and **resoluteness** to ensure that migrating species can continue their journeying a while longer.

**Crisis**= disaster, catastrophe, trouble, hard times 'kraɪsɪs

**Threaten**= endanger, put smb/smt at risk 'θreɪn

**Choke off**= to prevent something from happening tʃəʊk ɒf

**Conservation**= protection, preservation, safeguarding kənseɪ'veɪʃən

**Agency**= group, organization 'eɪdʒənsi

**Preserve**= maintain, protect, conserve prɪ'zɜ:v

**Recognize**= identify, know 'rekəɡnaɪz

**Path**= way, route pɑ:θ

**Jurisdiction**= rules, legal decisions dʒʊərəs'dɪkʃən

**Wisdom**= understanding, intelligence, knowledge 'wɪzdəm

**Resoluteness**= determination, perseverance 'rezəlu:təs

## READING PASSAGE 3



# Preface to “How the other half thinks: Adventures in mathematical reasoning”

A

**O**ccasionally, in some difficult musical

**compositions**, there are beautiful, but easy parts - parts so simple a beginner could play them. So it is with mathematics as well. There are some discoveries in **advanced** mathematics that do not depend on **specialized** knowledge, not even on algebra, **geometry**, or trigonometry. Instead they may involve, at most, a little **arithmetic**, such as ‘the sum of two **odd numbers** is **even**’, and common sense. Each of the eight chapters in this book **illustrates** this phenomenon. Anyone can understand every step in the reasoning. The thinking in each chapter uses at most only **elementary arithmetic**, and sometimes not even that. Thus all readers will have the chance to

**Reasoning**= the process of thinking about something in order to make a decision  
'ri:zəniŋ

**Composition**= work of art, piece of music, etc kəm'pə'ziʃən

**Advanced**= higher # foundation, elementary  
əd'vɑ:nst

**Specialized**= designed or developed for a particular purpose # generalized 'speʃ.əl.aɪzd

**Geometry**= the study in mathematics of the angles and shapes dʒi'ɒmətri

**Arithmetic**= the science of numbers involving adding, multiplying etc (*arithm*=count, number .i.e *logarithm*, *antilogarithm*) ə'riθməti:k

**Odd number**= a number that cannot be divided exactly by two, for example 1,3, 5,7 etc. # **even number** ɒd 'nʌmbə

**Illustrate**= demonstrate, show, describe  
'iləstreɪt

**Elementary**= basic, simple, straightforward, uncomplicated elə'mentəri

**Arithmetic**= the science of numbers involving adding, multiplying etc ə'riθməti:k

**participate in** a mathematical experience, to **appreciate** the beauty of mathematics, and to become familiar with its logical, yet **intuitive**, style of thinking.

B

One of my purposes in writing this book is to give readers who haven't had the opportunity to see and enjoy real mathematics the chance to appreciate the mathematical way of thinking. I want to **reveal** not only some of the **fascinating** discoveries, but, more importantly, the reasoning behind them. In that respect, this book differs from most books on mathematics written for the general public. Some present the lives of colorful mathematicians. Others describe important **applications** of mathematics. Yet others go into mathematical **procedures**, but **assume** that the reader is **adept** in using algebra.

C

I hope this book will help **bridge** that **notorious gap** that **separates** the two cultures: the humanities and the sciences, or should I say the right brain (intuitive) and the left brain (analytical, numerical). As the chapters will illustrate, mathematics is not restricted to the analytical and numerical; **intuition** plays a significant role. The **alleged** gap can be narrowed or completely **overcome** by anyone, in part because each of us is far from using the full **capacity** of either side of the brain. To illustrate our human **potential**, I **cite** a structural engineer who is an artist, an electrical engineer who is an opera singer, an opera singer who published mathematical research, and a mathematician who publishes short stories.

D

Other scientists have written books to explain their fields to non-scientists, but have necessarily had to **omit** the mathematics, although it provides the **foundation** of their theories. The reader must remain a **tantalized spectator** rather than an involved participant, since the **appropriate** language for describing the details in much of science is

**Participate in (v)**= take part in *pa:'tisəpɜrt ɪn*  
**Participant (n)**= a person who takes part in or becomes involved in a particular activity  
*pa:'tisəpɜnt*

**Appreciate**= value, respect, cherish, treasure, think highly of *ə'pri:ʃieɪt*

**Intuitive**= instinctive, natural *ɪn'tju:ətɪv*

**Reveal**= bring to light, show, point out  
*ri'vi:l*

**Fascinating**= interesting, attractive  
 # boring, dull *'fæsəneɪtɪŋ*

**Application**= implementation, use, utilization *æplɪ'keɪʃən*

**Procedure**= process, method, practice  
*prə'si:dʒə*

**Assume**= presume, suppose, guess...  
*ə'sju:m*

**Adept**= skillful, expert, proficient, gifted, talented # Inept *'ædept*

**Bridge a gap**= to make the difference between them smaller *brɪdʒ ə gæp*

**Notorious**= infamous *nəʊ'tɔ:riəs*

**Separate**= divide, split up, break up  
*'sepəreɪt*

**Intuition**= the ability to understand or know something because of a feeling rather than by considering the facts.  
*ɪn'tju:'ɪʃən*

**Alleged**= unproven, suspected  
 # confirmed *ə'ledʒd*

**Overcome**= defeat, control, get over  
*əʊvə'kʌm*

**Capacity**= ability *kə'pæsəti*

**Potential**= capability, capacity, ability  
*pə'tenʃəl*

**Cite**= name, mention, refer to *sart*

**Omit**= neglect, ignore, forget *əʊ'mɪt*

**Foundation**= base, basis *faʊn'deɪʃən*

**Tantalize**= to excite or attract someone by offering or suggesting something that is unlikely to be provided or is not enough  
*'tæntəl-aɪz*

**Spectator**= viewer, watcher, observer  
*spek'teɪtə*

**Appropriate**= suitable, acceptable *ə'preʊpri-ət*

mathematics, whether the subject is expanding universe, subatomic particles, or chromosomes. Though the broad outline of a scientific theory can be **sketched** intuitively, when a part of the physical universe is finally understood, its **description** often looks like a page in a mathematics text.

E

Still, the non-mathematical reader can go far in understanding mathematical reasoning. This book presents the details that illustrate the mathematical style of thinking, which involves **sustained**, step-by-step **analysis**, **experiments**, and **insights**. You will turn these pages much more slowly than when reading a novel or a newspaper. It may help to have a pencil and paper ready to check **claims** and **carry out** experiments.

F. As I wrote, I kept in mind two types of readers: those who enjoyed mathematics until they were turned off by an unpleasant episode, usually around fifth grade, and mathematics **aficionados**, who will find much that is new throughout the book. This book also serves readers who simply want to **sharpen** their **analytical** skills. Many careers, such as law and medicine, require extended, **precise** analysis. Each chapter offers practice in following a sustained and closely argued line of thought. That mathematics can develop this skill is shown by these two **testimonials**:

G

A physician wrote, the **discipline** of analytical thought processes [in mathematics] prepared me extremely well for medical school. In medicine one is faced with a problem which must be thoroughly analyzed before a solution can be found. The process is similar to doing mathematics.'

A lawyer made the same point, "Although I had no **background** in law - not even one political science course — I did well at one of the best law schools. I **attribute** much of my success there to having learned, through the study of mathematics, and, in particular, **theorems**, how to analyze principles. Lawyers who

**Sketch**= draft, outline sketʃ

**Description**= a piece of writing or speech that gives details about smb/smt dɪ'skrɪpʃən

**Sustained**= continuous, ongoing, constant sə'steɪnd

**Analysis**= study, investigation, examination ə'nælɪsɪs

**Experiment**= test ɪk'sperəmənt

**Insight**= a clear, deep understanding of a complicated problem or situation 'ɪnsaɪt

**Claim**= a statement that something is true, even though it has not been proved kleɪm

**Carry out**= do, conduct 'kæri 'aʊt

**Aficionado**= enthusiast, fan, admirer ə,fɪʃə'nɑ:dəʊ

**Sharpen**= enhance, improve 'ʃɑ:pən

**Analytical**= logical, systematic, scientific, rational ænəl'ɪtɪkəl

**Precise**= exact, accurate prɪ'saɪs

**Testimonials**= statements, confirmations testi'məʊniəlz

**Discipline**= an area of knowledge or teaching, especially one such as history, chemistry, mathematics etc that is studied at a university 'dɪsɪplɪn

**Background**= experience 'bækgraʊnd

**Attribute smt to smb/smt**= to believe or say that a situation or event is caused by something ə'trɪbjʊ:t 'sʌmθɪŋ tə 'sʌmbədi/ 'sʌmθɪŋ

**Theorem**= a statement that can be shown to be true 'θɪərəm

|                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>have studied mathematics can master the legal principles in a way that most others cannot.'</p> <p>I hope you will share my <b>delight</b> in watching as simple, even <b>naive</b>, questions lead to <u><b>remarkable</b></u> solutions and <b>purely</b> theoretical discoveries find <b>unanticipated</b> applications.</p> | <p><b>Delight</b>= enjoyment, pleasure, happiness, joy <span>di'laɪt</span></p> <p><b>Naive</b>= simple, inexperienced, immature<br/># sophisticated <span>naɪ'i:v</span></p> <p><b>Remarkable</b>= extraordinary, amazing, outstanding, significant, incredible<br/># ordinary <span>rɪ'mɑ:kəbəl</span></p> <p><b>Purely</b>= completely and only <span>'pjʊəli</span></p> <p><b>Unanticipated</b>= unexpected, surprising, unforeseen <span>ʌn.æn'tɪs.ɪ.peɪ.tɪd</span></p> |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## TEST 4

### READING PASSAGE 1



**T**o **biomedical** researchers all over the world, **twins** offer a **precious** opportunity to **untangle** the influence of genes and the environment - of nature and **nurture**. Because **identical** **twins** come from a single **fertilized** egg that **splits** into two, they share **virtually** the same genetic code. Any differences between them - one twin having younger looking skin, for example - must be due to environmental factors such as less time spent in the sun.

Alternatively, by comparing the experiences of identical

**Biomedical**= relating to how biology affects medicine *baɪəʊ'medɪkl*

**Twin**= one of two children born at the same time to the same mother *twɪn*

**Precious**= valuable, cherished, treasured *'preʃəs*

**Untangle**= understand or explain something that is mysterious or complicated *ʌn'tæŋɡəl*

**Nurture**= the education and care that you are given as a child, and the way it affects your later development and attitudes *'nɜːtʃə*

**Identical twin**= one of a pair of brothers or sisters born at the same time, who develop from the same egg and look almost exactly alike *aɪ'dentɪkəl twɪn*

**Fertilize** = to cause an egg or seed to start to develop into a new young animal or plant by joining it with a male cell *'fɜːtəlaɪz*

**Split**= divide, break # merge *splɪt*

**Virtually**= almost, nearly *'vɜːtʃuəli*

twins with those of **fraternal** twins, who come from separate eggs and share on average half their DNA, researchers can **quantify** the **extent** to which our genes affect our lives. If identical twins are more similar to each other with respect to an **ailment** than fraternal twins are, then **vulnerability** to the disease must **be rooted** at least in part in **heredity**.

These two lines of research - studying the differences between identical twins to **pinpoint** the influence of environment, and comparing identical twins with fraternal ones to measure the role of **inheritance** - have been **crucial** to understanding the **interplay** of nature and nurture in determining our personalities, behavior, and vulnerability to disease.

The idea of using twins to measure the influence of heredity dates back to 1875, when the English scientist Francis Galton first suggested the approach (and **coined the phrase** 'nature and nurture'). But twin studies took a surprising **twist** in the 1980s, with the arrival of studies into identical twins who had been separated at birth and **reunited** as adults. Over two decades 137 sets of twins eventually visited Thomas Bouchard's lab in what became known as the Minnesota Study of Twins Reared Apart. Numerous tests were **carried out** on the twins, and they were each asked more than 15,000 questions.

Bouchard and his colleagues used this mountain of data to identify how far twins were affected by their **genetic makeup**. The key to their approach was a **statistical** concept called heritability. In broad terms, the heritability of a **trait** measures the extent to which differences among members of a population can be explained by differences in their genetics. And wherever Bouchard and other scientists looked, it seemed, they found the **invisible** hand of genetic influence helping to shape our lives.

Lately, however, twin studies have helped lead scientists to a **radical** new conclusion: that nature and

**Fraternal**=connected with the relationship that exists between brothers frə'tɜːnl  
**Quantify**= count, calculate, measure 'kwɒntɪfaɪ  
**Extent**= degree, level ɪk'stɛnt  
**Ailment**= illness, disease 'eɪlmənt  
**Vulnerability**= the state of being easily affected by something vʌl.nər.ə'bɪl.ə.ti  
**Be rooted in something** = to be based on something or caused by something bi 'ruːtɪd ɪn 'sʌmθɪŋ  
**Heredity**= the process by which mental and physical qualities are passed from a parent to a child before the child is born.  
*(here=cling, stick .i.e adhere, coherence)*  
 hə'redəti

**Pinpoint**= identify 'pɪnpɔɪnt  
**Inheritance**= money or objects that someone gives you when they die ɪn 'herɪtəns  
**Crucial**= extremely important or necessary 'kruːʃəl  
**Interplay**= relationship, interaction  
*(inter- = among, between, i.e: intersection, international)* 'ɪntəpleɪ

**Coin the phrase**= invent a term kɔɪn ðə freɪz

**Twist** = change twɪst

**Reunite**= bring back together, come together riːjuː'naɪt

**Carry out**= do, implement 'kæri 'aʊt

**Genetic makeup**= genetic combination dʒɪ'netɪk 'meɪkʌp

**Statistical**= numerical stə'tɪstɪkəl

**Trait**= a particular quality in your personality treɪt

**Invisible** = impossible to see ɪn'vɪzəbəl

**Radical**= fundamental, basic, essential 'rædɪkəl

nurture are not the only **elemental** forces at work. According to a recent field called **epigenetics**, there is a third factor also in play, one that in some cases serves as a bridge between the environment and our genes, and in others operates on its own to shape who we are.

Epigenetic processes are chemical reactions **tied** to neither nature nor nurture but **representing** what researchers have called a 'third **component**'. These reactions influence how our genetic code is expressed: how each gene is strengthened or weakened, even turned on or off, to build our bones, brains and all the other parts of our bodies.

If you think of our DNA as an **immense** piano keyboard and our genes as the keys - each key symbolizing a **segment** of DNA responsible for a particular note, or **trait**, and all the keys combining to make us who we are - then epigenetic processes **determine** when and how each key can be struck, changing the **tune** being played.

One way the study of epigenetics is **revolutionizing** our understanding of biology is by **revealing** a **mechanism** by which the environment directly impacts on genes. Studies of animals, for example, have shown that when a rat experiences stress during **pregnancy**, it can cause epigenetic changes in a **fetus** that lead to behavioral problems as the rodent grows up. Other epigenetic processes appear to **occur** **randomly**, while others are normal, such as those that guide embryonic cells as they become heart, brain, or liver cells, for example.

Geneticist Danielle Reed has worked with many twins over the years and thought deeply about what twin studies have taught us. 'It's very clear when you look at twins that much of what they share is **hardwired**,' she says. 'Many things about them are absolutely the same and **unalterable**. But it's also clear, when you get to know them, that other things about them are different. Epigenetics is the origin of a lot of those differences, in my view.'

Reed credits Thomas Bouchard's work for today's **surge** in twin studies. 'He was the **trailblazer**,' she

**Elemental**= basic, fundamental *elə'mentl*

**Epigenetics**= the scientific study of changes in the behaviour of genes that are caused by things that a living thing experiences *epədʒə'netiks*

**Tied**= linked, joined, related *taɪd*

**Represent**= stand for, symbolize *reprɪ'zent*

**Component**= part, factor, element *kəm'pəʊnənt*

**Immense**=extremely large, enormous *ɪ'mens*

**Segment**= section, part, piece, sector *'segment*

**Trait**= characteristic, feature *treɪt*

**Determine**= decide *dɪ'tɜ:mɪn*

**Tune**= melody *tju:n*

**Revolutionize**= develop, transform *revə'lʊ:ʃənəɪz*

**Reveal**= show, display, exhibit # conceal *rɪ'vi:l*

**Mechanism** = process, system, operation *'mekənɪzəm*

**Pregnancy**= when a woman has a baby growing inside her body *'pregnənsɪ*

**Fetus**= a baby or young animal before it is born *'fi:təs*

**Occur** = happen *ə'kɜ:*

**Randomly** = unexpectedly = in an unplanned way *'rændəmli*

**Hardwired**= if an attitude, way of behaving etc is hard-wired, it is a natural part of a person's character that they are born with and cannot change *hɑ:d'waɪəd*

**Unalterable**= unchangeable, fixed *ʌn'ɔ:ltərəbəl*

**Surge** = sudden and great increase *sɜ:dʒ*

**Trailblazer**= pioneer *'treɪl,bleɪzə*

says. 'We forget that 50 years ago things like heart disease were thought to be caused entirely by lifestyle. Schizophrenia was thought to be due to poor **mothering**. Twin studies have allowed us to be more **reflective** about what people are actually born with and what's caused by experience.'

Having said that, Reed adds, the latest work in epigenetics promises to **take our understanding even further**. 'What I like to say is that nature writes some things in pencil and some things in pen,' she says. 'Things written in pen you can't change. That's DNA. But things written in pencil you can. That's epigenetics. Now that we're actually able to look at the DNA and see where the pencil writings are, it's sort of a whole new world.'

**Mothering**= the process of caring for children in the way that a mother does.

'mʌðərɪŋ

**Reflective**= thinking carefully and quietly

rɪ'flektɪv

**Take our understanding even further** =

increase our knowledge 'teɪk 'aʊər

ʌndə'stændɪŋ i:vən 'fɜ:ðə

## READING PASSAGE 2



**T**hough we might think of film as an essentially **visual** experience, we really cannot afford to **underestimate** the importance of film sound. A meaningful sound track is often as **complicated** as the image on the screen, and is ultimately just as much the responsibility of the director. The entire sound track consists of three **essential ingredients**: the human voice, sound effects and music. These three tracks must be mixed and **balanced** so as to produce the necessary **emphases** which in turn create **desired** effects.

Topics which essentially refer to the three previously mentioned tracks are discussed below. They include **dialogue**, synchronous and asynchronous sound effects, and music.

Let us start with dialogue. As is the case with stage drama, dialogue serves to tell the story and expresses feelings and **motivations** of characters as well. Often with film **characterization** the audience **perceives**

**Visual** = relating to seeing 'vɪʒuəl  
**Underestimate**= underrate, undervalue  
 # overestimate ʌndər'estimeɪt  
**Complicated**= complex # simple, easy  
 'kɒm.plɪ.keɪ.tɪd  
**Essential**= necessary or needed ɪ'senʃəl  
**Ingredient**= element, factor, part  
 ɪn'ɡri:diənt  
**Balanced** = containing an equal amount  
 or number of similar things or people  
 'bæl.ənst  
**Emphasis**= special attention or  
 importance 'emfəsis  
**Desired** = expected, wanted # unwanted  
 dɪ'zaɪəd

**Dialogue**= conversation 'daɪələg  
**Motivation**= enthusiasm for doing  
 something məʊtə'veɪʃən  
**Characterization**= the way in which the  
 character of a real person or thing is  
 described kærəktəraɪ'zeɪʃən  
**Perceive**= recognize, become aware of  
 pə'si:v

little or no difference between the character and the actor. Thus, for example, the actor Humphrey Bogart is the character Sam Spade; film **personality** and life personality seem to **merge**. Perhaps this is because the very **texture** of a performer's voice supplies an **element** of character.

When voice textures fit the performer's **physiognomy** and **gestures**, a whole and very realistic **personal emerges**. The viewer sees not an actor working at his **craft**, but another human being **struggling** with life. It is interesting to note that how dialogue is used and the very amount of dialogue used varies widely among films. For example, in the highly successful science-fiction film 2001, little dialogue was evident, and most of it was **banal** and of little **intrinsic** interest. In this way the film-maker was able to **portray** what Thomas Sobotchack and Vivian Sobotchack call, in An Introduction to Film, the '**inadequacy** of human responses when compared with the **magnificent** technology created by man and the visual beauties of the universe'.

The comedy Bringing Up Baby, on the other hand, presents practically non-stop dialogue delivered at **breakneck** speed. This use of dialogue **underscores** not only the **dizzy** quality of the character played by Katherine Hepburn, but also the **absurdity** of the film itself and thus its humor. The audience is bounced from gag to gag and conversation to conversation; there is no time for audience **reflection**. The audience is **caught up in** a whirlwind of activity in simply managing to follow the **plot**. This film presents pure **escapism** - largely due to its **frenetic** dialogue.

Synchronous sound effects are those sounds which are synchronized or matched with what is viewed. For

**Personality**= traits, character p3:sə'næləti

**Merge**= combine, unite # separate, split m3:dʒ

**Texture**= the quality of something (eg. voice) that is rough or smooth or soft or hard 'tekstʃə

**Element**= a part of sth 'eləmənt

**Physiognomy**= the general appearance of a person's face fɪzi'ɒnəmi

**Gesture**= a movement of part of your body, especially your hands or head, to show what you mean or how you feel 'dʒestʃə

**Personal** = belonging or relating to one particular person 'p3:sənəl

**Emerge**= to appear by coming out of something or out from behind something i'm3:dʒ

**Craft**= skill, ability, technique kɹɑ:ft

**Struggle**= to experience difficulty and make a very great effort in order to do something 'strʌgl

**Banal**=trivial= ordinary and not interesting, because of a lack of new or different ideas. bə'na:l

**Intrinsic**= natural in'trɪnsɪk

**Portray**= to represent or describe someone or something in a painting, film, book, or other artistic work pɔ:'treɪ

**Inadequacy**= insufficiency, lack

# abundance in'ædəkwəsi

**Magnificent**= wonderful, outstanding, brilliant mæg'nɪfəsənt

**Breakneck**= fast, rapid, quick 'breɪknek

**Underscore**= emphasize, highlight, underline # ignore ʌndə'sko:

**Dizzy**= stupid and forgetful 'dɪzi

**Absurdity**= illogicality, ridiculousness, foolishness # logic, reasonableness əb'sɜ:d

**Reflection**=serious and careful thought rɪ'flekʃən

**be/get caught up in something**= to be or get involved in something, especially something bad bi/ 'get 'kɔ:t 'ʌp ɪn 'sʌmθɪŋ

**Plot**= story line plɒt

**Escapism** = activities or entertainment that help you forget about bad or boring things for a short time i'skeɪpɪzəm

**Frenetic**= frantic, fast and not very organized frə'netɪk

example, if the film portrays a character playing the piano, the sounds of the piano are projected. Synchronous sounds contribute to the realism of film and also help to create a particular atmosphere. For example, the 'click' of a door being opened may simply serve to **convince** the audience that the image portrayed is real, and the audience may only **subconsciously** note the expected sound. However, if the 'click' of an opening door is part of an **ominous** action such as a **burglary**, the sound mixer may call attention to the 'click' with an increase in volume; this helps to **engage** the audience in a moment of **suspense**.

Asynchronous sound effects, on the other hand, are not matched with a visible source of the sound on screen. Such sounds are included so as to provide an **appropriate** emotional **nuance**, and they may also add to the realism of the film. For example, a film-maker might **opt** to include the background sound of an ambulance's siren while the foreground sound and image portrays an arguing couple. The asynchronous ambulance siren underscores the psychic **injury** incurred in the argument; at the same time the noise of the siren adds to the realism of the film by acknowledging the film's city setting. We are probably all familiar with background music in films, which has become so **ubiquitous** as to be **noticeable** in its **absence**. We are aware that it is used to add **emotion** and **rhythm**. Usually not meant to be noticeable, it often provides a tone or an emotional attitude toward the story and /or the characters depicted. In addition, background music often **foreshadows** a change in mood. For example, **dissonant** music may be used in film to **indicate** an approaching (but not yet visible) **menace** or disaster. Background music may **aid** viewer understanding by linking scenes. For example, a particular musical theme **associated with** an individual character or situation may be repeated at various points in a film in order to remind the audience of **salient motifs** or ideas.

**Convince**= persuade kən'vɪns

**Subconsciously**= unintentional  
# consciously səb'kɒn.jəs.li

**Ominous**=suggesting that something unpleasant is likely to happen 'ɒmɪnəs  
**Burglary**= breaking and entering, theft, robbery 'bɜ:gləri

**Engage somebody in something**= to attract someone's attention and keep them interested ɪn'geɪdʒ 'sʌmbədi ɪn 'sʌmθɪŋ

**Suspense**= tension, uncertainty, doubt, anxiety, nervousness sə'spens

**Appropriate**=suitable or right for a particular situation or occasion ə'prəʊpri-ət

**Nuance**= a very slight, hardly noticeable difference in manner, colour, meaning 'nju:ɑ:ns

**Opt**= to make a choice, especially of one thing or possibility instead of others ɒpt

**Injury**= physical harm or damage to someone's body caused by an accident or an attack 'ɪndʒəri

**Ubiquitous** = popular, common, pervasive, prevalent # rare ju:'bɪkwɪtəs

**Noticeable**= clear, obvious 'nəʊtəsəbəl

**Absence**= nonexistence, lack 'æbsəns

**Emotion**= feeling ɪ'məʊʃən

**Rhythm**= beat 'rɪðəm

**Foreshadow**= anticipate fə:'ʃædəʊ

**Dissonant** = lacking harmony 'dɪs.ən.ənt

**Indicate**=to show, point, or make clear in another way 'ɪndɪkeɪt

**Menace**= threat, danger 'menɪs

**Aid**= help or support eɪd

**Associate with**= link with, connect with ə'səʊsɪeɪt wɪð

**Salient**= important, main, outstanding, noticeable, obvious, remarkable 'seɪliənt

**Motif**= pattern məʊ'ti:f

Film sound **comprises** **conventions** and **innovations**. We have come to expect an **acceleration** of music during car chases and creaky doors in horror films. Yet, it is important to note as well that sound is often brilliantly **conceived**. The effects of sound are often largely **subtle** and often are noted by only our **subconscious** minds. We need to **foster** an awareness of film sound as well as film space **so as to** truly **appreciate** an art form that sprang to life during the twentieth century - the modern film.

**Comprise**= include, contain, consist of kəm'praɪz  
**Convention**= rule, standard, principle kən'venʃən  
**Innovation**= improvement, modernization ɪnə'veɪʃən  
**Acceleration**= speeding up, quickening, increase in speed ək'selə'reɪʃən  
**Conceive**= think, see, perceive, grasp, appreciate, apprehend kən'si:v  
**Subtle**= delicate, indirect # obvious 'sʌtl  
**Foster**= promote, encourage 'fɒstə  
**So as to**= in order to, in an attempt to 'səʊ əz tu:  
**Subconscious**=connected with feelings that influence your behaviour even though you are not aware of them sʌb'kɒn.ʃəs  
**Appreciate** = acknowledge, recognize, be aware of ə'pri:jeɪt



# This Marvellous Invention

**O**f all mankind's **manifold** creations, language

Λη'pær.əl.eld

B

But language is **foremost** not just because it came first. In its own right it is a tool of **extraordinary sophistication**, yet based on an idea of **ingenious simplicity**: 'this **marvellous** invention of composing out of twenty-five or thirty sounds that **infinite** variety of expressions which, whilst having in themselves no likeness to what is in our mind, allow us to **disclose** to others its whole secret, and to make known to those who cannot **penetrate** it all that we imagine, and all the various **stirrings of our soul**' This was how, in 1660, the **renowned** French grammarians of the Port-Royal abbey near Versailles **distilled** the **essence** of language, and no one since has celebrated more eloquently the **magnitude** of its achievement. Even so, there is just one **flaw** in all these **hymns** of praise, for the **homage** to languages unique **accomplishment** **conceals** a simple yet **critical** **incongruity**. Language is mankind's greatest invention - except, of course, that it was never invented. This apparent **paradox** is at the core of our fascination with language, and it holds many of its secrets.

C

Language often seems so skillfully drafted that one can hardly imagine it as anything other than the perfected **handiwork** of a master **craftsman**. How else could this **instrument** make so much out of barely three dozen measly morsels of sound? In themselves, these **configurations** of mouth p,f,b,v,t,d,k,g,sh,a,e and so on - **amount to** nothing more than a few **haphazard** spits and splutters, random noises with no meaning, no ability to express, no power to explain. But run them through the cogs and wheels of the language machine,

**Foremost**=most important or best; leading  
'fɔ:məʊst

**Extraordinary**=very unusual, special, unexpected, or strange  
ɪk'strɔ:dənəri

**Sophistication**= complexity  
sə,fɪs.tɪ'keɪ.ʃən

**Ingenious**=very suitable for a particular purpose and resulting from clever new ideas  
ɪn'dʒi:ni.əs

**Simplicity**= easiness # complexity  
sɪm'plɪsəti

**Marvellous**=extremely good; wonderful  
'mɑ:veləs

**Infinite**=very great; impossible to measure  
'ɪnfənət

**Disclose**= reveal, make known # hide  
dɪs'kləʊz

**Penetrate**= understand, grasp, perceive  
'penetret

**Stirring of our soul**= the arousing of deep emotion  
'stɜ:ɪŋ əv 'aʊə səʊl

**Renowned**= famous, well-known  
rɪ'naʊnd

**Distill**= to get the main ideas or facts from a much larger amount of information  
dɪ'stɪl

**Essence**= the basic or most important idea or quality of something  
'esəns

**Magnitude**=the large size or importance of something  
'mæɡnɪtju:d

**Flaw**= fault, error  
flɔ:

**Hymn**= a song of praise  
hɪm

**Homage**= respect, honor  
'hɒmɪdʒ

**Accomplishment**= achievement  
ə'kʌm.plɪʃ.mənt

**Conceal**= hide, cover up  
kən'si:l

**Critical**=extremely important because a future situation will be affected by it  
'krɪtɪkəl

**Incongruity**= strangeness, inappropriateness  
ɪn'kɒn'ɡru:əti

**Paradox**= inconsistency, contradiction = a situation or statement that seems impossible or is difficult to understand because it contains two opposite facts or characteristics  
'pærədɒks

**Handiwork**= something done skillfully with the hands  
'hændɪwɜ:k

**Craftsman**= someone who is very skilled at a particular craft  
'krɑ:ftsmən

**Instrument**= tool, gadget  
'ɪnstɾəmənt

**Configuration**= the shape or arrangement of the parts of something (*fig=form, shape .i.e figure, configure*)  
kən'fɪɡə'reɪʃən

**Amount to something**= be the same as  
ə'maʊnt tu 'sʌmθɪŋ

**Haphazard**= random, disorganized  
hæp'hæzəd

let it arrange them in some very special orders, and there is nothing that these **meaningless** streams of air cannot do: from sighing the **interminable** boredom of existence to **unravelling** the fundamental order of the universe.

## D

The most extraordinary thing about language, however, is that one doesn't have to be a genius to set its wheels in **motion**. The language machine allows just about everybody from pre-modern **foragers** in the subtropical savannah, to post-modern philosophers in the suburban sprawl - to tie these meaningless sounds together into an **infinite** variety of **subtle** senses, and all apparently without the slightest **exertion**. Yet it is precisely this **deceptive ease** which makes language a **victim of its own success**, since in everyday life its **triumphs** are usually **taken for granted**. The wheels of language run so smoothly that one rarely bothers to stop and think about all the resourcefulness and **expertise** that must have gone into making it tick. Language **conceals** art.

## E

Often, it is only the **estrangement** of foreign tongues, with their many **exotic** and **outlandish** features, that brings home the wonder of languages design. One of the **showiest stunts** that some languages can **pull off** is an ability to build up words of breath-breaking length, and thus express in one word what English takes a whole sentence to say. The Turkish word *çehirliliçtiremediklerimizdensiniz*, to take one example, means nothing less than 'you are one of those whom we can't turn into a town-dweller'. (In case you were wondering, this **monstrosity** really is one word, not merely many different words **squashed** together - most of its **components** cannot even stand up on their own.)

## F

**Meaningless**= worthless # meaningful  
'mi:nɪŋləs  
**Interminable**= endless= long and boring  
ɪn'tɜ:mənəbəl  
**Unravel**= to understand or explain something that is mysterious or complicated ʌn'rævəl

**Motion**= the act or process of moving or the way something moves 'məʊʃən

**Foragers**= hunters 'fɔ:ɪdʒ.ərs

**Infinite**= never-ending, unlimited 'ɪnfənət

**Subtle**= not easy to notice or understand unless you pay careful attention 'sʌtl

**Exertion**= a lot of mental and physical effort ɪg'zɜ:ʃən

**Deceptive**= misleading, false dɪ'septɪv

**Ease**= effortlessness, no difficulty i:z

**Be a victim of your own success**= to be badly affected by some unexpected results of being very successful bi ə 'vɪktɪm əv jər əʊn sək'ses

**Triumph**= victory, win, conquest, success 'traɪəmf

**Take for granted**= to believe something is true without first making sure that it is 'teɪk fə 'grɑ:ntɪd

**Expertise**=expert knowledge or skill in a particular subject, activity or job ekspɜ:'ti:z

**Conceal**=to hide somebody/something kən'si:l

**Estrangement** = separation, division used to describe one of several parts of which something is made ɪ'streɪndʒd

**Exotic**= unusual, out of the ordinary, strange # familiar, conventional ɪg'zɒtɪk

**Outlandish**= strange, unusual, weird ʌʊt'lændɪʃ

**Showy stunt**= something that is done to attract people's attention, especially in advertising or politics. 'ʃəʊ stʌnt

**Pull off**= succeed, do well pul əf

**Monstrosity** = something that is very ugly and usually large mɒn'strɒsəti

**Squash** = squeeze = to push something into a small space skwɒʃ

**component**=used to describe one of several parts of which something is made kəm'pəʊnənt

And if that sounds like some one-off freak, then consider Sumerian, the language spoken on the banks of the Euphrates some 5,000 years ago by the people who invented writing and thus **enabled** the **documentation** of history. A Sumerian word like *munintuma'a* ('when he had made it suitable for her') might seem rather **trim** compared to the Turkish colossus above. What is so **impressive** about it, however, is not its lengthiness but rather the **reverse** - the **thrifty compactness** of its construction. The word is made up of different slots, each **corresponding to** a particular portion of meaning. This **sleek** design allows single sounds to **convey** useful information, and in fact even the **absence** of a sound has been enlisted to express something specific. If you were to ask which bit in the Sumerian word corresponds to the pronoun 'it' in the English translation 'when he had made it suitable for her', then the answer would have to be nothing. Mind you, a very particular kind of nothing: the nothing that stands in the empty slot in the middle. The technology is so **fine-tuned** then that even a non-sound, when carefully placed in a particular position, has been invested with a specific function. Who could possibly have **come up with** such a **nifty contraption**?

**Enable**= allow, make possible # prevent  
ɪˈneɪbəl

**Documentation**= records dɒkjəməntəˈteɪʃən

**Trim**= neat, shorten trɪm

**Impressive**= extraordinary, remarkable  
# unimpressive, ordinary ɪmˈpresɪv

**Reverse**= opposite, contrary rɪˈvɜːs

**Thrifty**= careful about spending money and not wasting things ˈθrɪfti

**Compactness**= neatness = the fact that everything fits neatly into a little space  
kəmˈpækt.nəs

**correspond to** = to be very similar to or the same as something else  
(cor-= together, with) kɒrɪˈspɒnd tuː

**Sleek**= smooth sliːk

**Convey**= express kənˈveɪ

**Absence**= lack, non-existence  
# presence, attendance ˈæbsəns

**Fine-tune** = to make very small changes to something such as a machine, system, or plan, so that it works as well as possible faɪnˈtjuːn

**Come up with**= to think of an idea, find out, create ˈkʌm ˈʌp wɪð

**Nifty**= ingenious, clever, effective ˈnɪfti

**Contraption**= a piece of equipment or machinery kənˈtræpʃən

## List of root words

| No. | Root, prefix, suffix | Meaning      | Examples                                                                                                                                                                                                                                                                                                                                                   |
|-----|----------------------|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.  | <b>urb</b>           | City         | <p><b>urban</b>= city, inner-city, metropolitan, town</p> <p><b>suburb</b>= an area on the edge of a large town or city where people who work in the town or city often live</p>                                                                                                                                                                           |
| 2.  | <b>dem-</b>          | People       | <p><b>demographic</b>= relating to the population and groups of people in it</p> <p><b>democracy</b>= the belief in freedom and equality between people, or a system of government based on this belief</p>                                                                                                                                                |
| 3.  | <b>re-</b>           | do again     | <p><b>renewal</b>= the <u>act</u> or <u>process</u> of making changes to something in <u>order</u> to <u>improve</u> it so that it becomes more <u>successful</u></p> <p><b>reverse</b>= to change something, such as a decision, judgment, or process so that it is the opposite of what it was before</p> <p><b>re-routing</b>= change the direction</p> |
| 4.  | <b>de-</b>           | reduce, down | <p><b>despoil</b>=damage, spoil, ruin</p> <p><b>deplete</b>= to reduce something in size or amount, especially supplies of energy, money, etc</p> <p><b>deforestation</b>= the cutting or burning down of all the trees in an area</p>                                                                                                                     |
| 5.  | <b>semi-</b>         | Half         | <p><b>semi-arid</b>= have little rain but is not completely dry</p>                                                                                                                                                                                                                                                                                        |

|     |                   |                  |                                                                                                                                                                                                   |
|-----|-------------------|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 6.  | <b>vol, volut</b> | change, turn     | <b>evolve</b> = <i>develop gradually</i><br><br><b>revolution</b> = <i>a movement in a circle or curve around a central point</i>                                                                 |
| 7.  | <b>optim</b>      | Best             | <b>optimum</b> = <i>most favorable, best, greatest</i>                                                                                                                                            |
| 8.  | <b>herb</b>       | Grass            | <b>herbicide</b> = <i>a substance used to kill unwanted plants</i><br><br><b>herbal</b> = <i>relating to or made from herbs</i>                                                                   |
| 9.  | <b>cid</b>        | Kill             | <b>pesticide</b> = <i>a chemical substance used to kill harmful insects, small animals, wild plants, and other unwanted organisms</i>                                                             |
| 10. | <b>inter-</b>     | among, between   | <b>interplay</b> = <i>relationship, interaction</i><br><br><b>intersection</b> = <i>the place where two or more roads join or cross each other</i>                                                |
| 11. | <b>ed</b>         | Eat              | <b>edible</b> = <i>suitable or safe for eating</i>                                                                                                                                                |
| 12. | <b>tract</b>      | pull, draw       | <b>tractor</b> = <i>a strong vehicle with large wheels, used for pulling farm machinery</i>                                                                                                       |
| 13. | <b>ex-</b>        | Out              | <b>expose</b> = <i>make something uncovered or <u>hidden</u> <u>able</u> to be <u>seen</u></i><br><br><b>exterior</b> = <i>the outside part of sth</i><br><br><b>exclude</b> = <i>not include</i> |
| 14. | <b>vari</b>       | Vary             | <b>variation</b> = <i>something that is done in a way that is different from the way it is usually done</i><br><br><b>variety</b> = <i>a different type of something</i>                          |
| 15. | <b>nav</b>        | relating to ship | <b>navigability</b> = <i>the <u>degree</u> to which an area of water is <u>deep</u>, <u>wide</u>,</i>                                                                                             |

|     |              |           |                                                                                                                                                                                                                     |
|-----|--------------|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     |              |           | <p>or <u>safe</u> enough for a <u>boat</u> to go through</p> <p><b>naval</b>= belonging to a country's navy, or relating to military ships</p>                                                                      |
| 16. | <b>memor</b> | Remember  | <p><b>commemoration</b>= something that makes you remember and respect someone important or an important event in the past</p> <p><b>memorable</b>= likely to be remembered or worth remembering</p>                |
| 17. | <b>mit</b>   | Send      | <p><b>submit</b>= <u>offer</u>, propose, suggest</p> <p><b>emit</b>= send out gas, heat, light, sound, etc</p>                                                                                                      |
| 18. | <b>mono</b>  | only, one | <p><b>monorail</b>= a railway system that uses a single rail, usually high above the ground</p> <p><b>monopoly</b>= complete control of something, especially an area of business, so that others have no share</p> |
| 19. | <b>hydr</b>  | Water     | <b>hydraulic</b> = operated by or involving the pressure of water or some other liquid                                                                                                                              |
| 20. | <b>orb</b>   | Circle    | <b>orbit</b> = to move in a curved path around a much larger object                                                                                                                                                 |
| 21. | <b>aqu-</b>  | Water     | <p><b>aqueduct</b>= a structure like a bridge, that carries water across a river or valley</p> <p><b>aquarium</b>= a glass container in which fish and other water creatures can be kept</p>                        |
| 22. | <b>lev</b>   | Rise      | <b>elevate</b> = raise, lift, make higher                                                                                                                                                                           |

|     |              |                             |                                                                                                                                                                                                                                         |
|-----|--------------|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     |              |                             | <b>elevation</b> = <i>the height of a place above the level of the sea</i>                                                                                                                                                              |
| 23. | <b>geo-</b>  | of or relating to the earth | <b>geo-engineering</b> = <i>the study of finding ways to change the earth's atmosphere in order to reduce global warming</i>                                                                                                            |
| 24. | <b>equ-</b>  | Equal                       | <b>equivalent</b> = <i>parallel, similar</i><br><b>equilibrium</b> = <i>a state of balance</i>                                                                                                                                          |
| 25. | <b>loc</b>   | Place                       | <b>localize</b> = <i>to <u>limit</u> smth to a particular <u>area</u></i><br><b>relocation</b> = <i>the process of moving to a different place to work, or of moving employees to a different place to work</i>                         |
| 26. | <b>min-</b>  | less, smaller               | <b>minute</b> = <i>tiny, little, small</i><br><b>minority</b> = <i>a smaller number/part</i>                                                                                                                                            |
| 27. | <b>-plen</b> | Full                        | <b>replenish</b> = <i>refill</i>                                                                                                                                                                                                        |
| 28. | <b>aer-</b>  | air, atmosphere             | <b>aerosol</b> = <i>a <u>metal container</u> in which liquids are kept under pressure and <u>forced</u> out in a <u>spray</u></i><br><b>aeroplane</b> = <i>a vehicle designed for air travel that has wings and one or more engines</i> |
| 29. | <b>serv</b>  | protet, save                | <b>preserve</b> = <i>protect</i><br><b>conservation</b> = <i>the protection of natural things such as animals, plants, forests etc, to prevent them from being spoiled or destroyed</i>                                                 |
| 30. | <b>over-</b> | too much                    | <b>overshoot</b> = <i>exceed, surpass</i><br><b>overladen</b> = <i>filled with too many people or things</i>                                                                                                                            |

|     |               |                             |                                                                                                                                                                                                                           |
|-----|---------------|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     |               |                             | <b>overrun</b> = <i>if a place is overrun by unwanted things or people, they spread over it in great numbers</i>                                                                                                          |
| 31. | <b>mis-</b>   | done wrongly, badly         | <b>mishandle</b> = <i>to treat something roughly, often causing damage</i><br><br><b>misbehavior</b> = <i>to behave badly</i>                                                                                             |
| 32. | <b>un-</b>    | the opposite of N/V/ADJ/ADV | <b>undisciplined</b> = <i>behaving in an uncontrolled way # disciplined, well behaved</i><br><br><b>unanticipated</b> = <i>not having been expected to happen</i>                                                         |
| 33. | <b>in-</b>    | not, un                     | <b>intact</b> = <i>unbroken. unharmed. undamaged # broken, damaged</i><br><br><b>infertile</b> = <i>unproductive # fertile</i>                                                                                            |
| 34. | <b>tact</b>   | Touch                       | <b>intact</b> = <i>untouched, unbroken, unharmed, undamaged, # broken, damaged</i><br><br><b>tactile</b> = <i>related to the sense of touch</i>                                                                           |
| 35. | <b>salv-</b>  | Save                        | <b>salvage</b> = <i>to save goods from damage or destruction, especially from a <u>ship</u> that has sunk</i>                                                                                                             |
| 36. | <b>civ</b>    | Citizens                    | <b>civilisation</b> = <b><i>human development</i></b><br><br><b>civic</b> = <i>of a town or city or the people who live in it</i>                                                                                         |
| 37. | <b>extra-</b> | Outer                       | <b>extraterrestrial</b> = <i>a creature that people think may exist on another planet</i><br><br><b>extracurricular</b> = <i>an extracurricular activity or subject is not part of the usual school or college course</i> |
| 38. | <b>terr-</b>  | Earth                       | <b>territory</b> = <i>a large area that has some local government</i><br><br><b>terrain</b> = <i>an area of land, when considering its natural features</i>                                                               |

|     |                           |                    |                                                                                                                                                                                                                                                                       |
|-----|---------------------------|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 39. | <b>archae-</b>            | Ancient            | <b>archaeological</b> = <i>belonging to the study of ancient societies by examining what remains of their buildings, graves, tools etc</i>                                                                                                                            |
| 40. | <b>neur(o)-</b>           | relating to nerves | <b>neuroscience</b> = <i>the scientific study of the nervous system and the brain</i><br><br><b>neuroaesthetics</b> = <i>a field of study in which researchers attempt to understand how the brain responds to art</i>                                                |
| 41. | <b>co-, cor-<br/>con-</b> | together, with     | <b>collaborator</b> = <i>coworkers, colleague, partner</i><br><br><b>correspond</b> = <i>to be very <u>similar</u> to or the same as something else</i><br><br><b>conjunction</b> = <i>combination</i>                                                                |
| 42. | <b>under-</b>             | not enough, below  | <b>underestimate</b> = <i>undervalue</i><br># <i>overestimate</i><br><br><b>underprivileged</b> = <i>without the money, possessions, education, opportunities, etc. that the average person has</i>                                                                   |
| 43. | <b>magn</b>               | great, large       | <b>magnitude</b> = <i>the large size or importance of something</i><br><br><b>magnificent</b> = <i>very good, beautiful, or deserving to be admired</i>                                                                                                               |
| 44. | <b>line-</b>              | Line               | <b>linear</b> = <i>following a straight line</i>                                                                                                                                                                                                                      |
| 45. | <b>arithm</b>             | count, number      | <b>arithmetic</b> = <i>the <u>science</u> of numbers involving adding, <u>multiplying</u> etc</i><br><br><b>logarithm</b> = <i>a number which shows how many times a particular number, called the base, has to be multiplied by itself to produce another number</i> |
| 46. | <b>mechan</b>             | Machine            | <b>mechanism</b> = <i>process, system, operation</i>                                                                                                                                                                                                                  |

|     |             |              |                                                                                                                                                                                                                                                     |
|-----|-------------|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 47. | <b>vis</b>  | See          | <p><b>visual</b> = <u>relating to seeing</u></p> <p><b>visit</b>= <i>to go to a place in order to look at it, or to a person in order to spend time with them</i></p>                                                                               |
| 48. | <b>fig</b>  | form, shape  | <p><b>figure</b>= <i>the symbol for a number or an amount expressed in numbers</i></p> <p><b>configuration</b>= <i>the shape or arrangement of the parts of something</i></p>                                                                       |
| 49. | <b>vers</b> | turn, change | <p><b>versatile</b>= <i>able to change easily from one activity to another or able to be used for many different purposes</i></p> <p><b>reverse</b>= <i>to change the direction, order, position, result, etc. of something to its opposite</i></p> |

## ARTICLES RELATED TO PASSAGES IN TEST 1

### TEST 1

#### PASSAGE 1: CROP- GROWING SKYSCRAPERS

##### ARTICLE 1: Growing Up: Skyscraper Farms Seen as a Way to Produce Food Locally-And Cut Greenhouse Emissions

Could growing crops on high-rise buildings feed growing urban populations, thereby sparing the need to **cultivate (Cam10 -cultivating)** more and more tracts of land?

Dear EarthTalk: What is "vertical farming" and how is it better for the environment?

—Jonathan Salzman, New York City

"Vertical farming" is a term coined by Columbia University professor of environmental health and microbiology Dickson Despommier to describe the concept of growing large amounts of food in urban high-rise buildings—or so-called "farmscrapers."

According to the vision first developed in 1999 by Despommier and his students, a 30-story building built on one city block and engineered to maximize year-round agricultural **yield (Cam11- yields)**—thanks largely to artificial lighting and advanced hydroponic and aeroponic growing techniques—could feed tens of thousands of people. Ideally the recipients of the bounty would live in the surrounding area, so as to avoid the transport costs and **carbon emissions (Cam 11)** associated with moving food hundreds if not thousands of miles to

**Cultivate** = Grow = Farm

**Yield**= the amount of profits, crops etc that something produces

**carbon emissions** = carbon dioxide that planes, cars, factories, etc. produce, thought to be harmful to the environment

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>consumers.</p> <p>“Each floor will have its own watering and nutrient monitoring systems,” Despommier <b>elaborated (Cam8)</b> to online magazine Miller-McCune.com, adding that every single plant’s health status and nutrient consumption would be tracked <b>by sensors (Cam 8-sensory, Cam 9)</b> that would help managers ward off diseases and increase yield without the need for the chemical fertilizers and pesticides so common in traditional outdoor agriculture.</p> <p>“Moreover, a gas chromatograph will tell us when to pick the plant by analyzing which flavenoids the produce contains,” Despommier said. “It’s very easy to do... These are all right-off-the-shelf technologies. The ability to construct a vertical farm exists now. We don’t have to make anything new.”</p> <p>With world population set to top nine billion by 2050 when 80 percent of us will live in cities, Despommier says vertical farming will be key to feeding an increasingly urbanized <b>human race (Cam 9,13)</b>. His Vertical Farm Project claims that a <b>vertical farm</b> on one acre of land can grow as much food as an outdoor farm on four to six acres. Also, vertical farms, being indoors, wouldn’t be subject to the vagaries of weather and pests.</p> <p>“The reason we need <b>vertical farming (Cam 11)</b> is that horizontal farming is failing,” Despommier told MSNBC, adding that if current practices don’t change soon, humanity will have to <b>devote (Cam9)</b> to agriculture an area bigger than Brazil to keep pace with global food demand. Another benefit of vertical farming is that former farmland could be returned to a natural state and even help fight <b>global warming (Cam 8,11)</b>. As agricultural land becomes forest and other green space, plants and trees there can store carbon dioxide while also serving as habitat for wildlife otherwise <b>displaced (Cam11- displace)</b> by development.</p> | <p><b>Elaborate</b>= intricate, complicated, complex...</p> <p><b>Sensors</b> = a device that reacts to heat, light, or movement</p> <p><b>human race</b> = all people, considered as a group</p> <p><b>vertical farming</b> = = an idea for a way of farming in which plants are grown or animals are kept in tall structures with many levels</p> <p><b>devote</b> = dedicate = give = offer</p> <p><b>global warming</b> = an increase in the temperature of the air and the oceans around the world</p> <p><b>Displace</b> = put out of place, move</p> |
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Vertical farming is not without critics, who argue that the practice would use huge amounts of electricity for the artificial lights and **machinery (Cam 9)** that would **facilitate (Cam 10)** year-round harvests. Bruce Bugbee, a Utah State University crop physiologist, believes that the power demands of vertical farming—growing crops requires about 100 times the amount of light as people working in office buildings—would make the practice too expensive compared to traditional farming where the primary input, sunlight, is free and **abundant (Cam 9)**. Proponents argue that vertical farms could produce their own power by tapping into local **renewable sources (Cam 9,11 renewable energy)** (solar, wind, tidal or geothermal) as well as by burning biomass from crop waste

**machinery** = machines, often large machines

**facilitate** = help = assist

**Abundant** = plentiful, rich, ample  
#scarce...

**Renewable** = Renewable forms of energy can be produced as quickly as they are used

## PASSAGE 2: THE FALKIRK WHEEL

### ARTICLES 2: 10 Most Amazing Engineering Achievements

#### 1. The International Space Station (ISS)

First on our list is a space age engineering **accomplishment (Cam 10,11)**. The ISS was launched in 1998 and has been orbiting Earth ever since. The station holds a **crew (Cam 12,13)** of 6 astronauts and is the largest **artificial (Cam 11,13)** object in orbit: it is 72.8 m (239 ft) long, 108.5 m (356 ft) wide and around 20 m (66 ft) in height. It is so big that it's possible to see the ISS from earth with the naked eye. The ISS is used for research in different fields of science and lately HD cameras that send live streaming of Earth over the internet were installed

#### 2. The Large Hadron Collider (LHC)

Still in the field of recent technology **achievements (cam 12,13)**, the Large Hadron Collider (LHC) is the largest particle collider in the world, built in the years 1998-2008 by CERN (the European Organization for Nuclear Research). It was built by thousands of scientists and engineers from hundreds of universities and from more than 100 countries. The LHC is a 27 km (17 mi) long tunnel, 175 metres (574 ft) deep in the ground beneath the borders of France and Switzerland. So what is the LHC used for? Simply put, it **enables (Cam 12,13)** scientists to **reproduce (Cam 9,10- reproduction)** the state of the universe (very) shortly after the Big Bang, and test **relevant (Cam 12,13)** theories. In 2013 an experiment in the LHC proved the existence of the Higgs boson **particle (Cam 11)**

#### 3. The Panama Canal

**Accomplishment=** achievement, success...

**Crew=** team, group

**Artificial=** synthetic, non-natural, man-made

**Achievement=** attainment, accomplishment

**Enable** = allow, make possible, permit, aid, support, facilitate, empower....

**reproduce** = to make a copy of something

**relevant** = related

**particle** = a very small piece of something

**Marvel=** wonder, awesome sight, amazing thing...

The Panama Canal is an engineering **marvel (Cam 8)** that truly shows how **mankind (Cam 11)** can overcome obstacles and not just live on Earth, but make changes on it's surface to better meet its needs. Built from 1881 till 1914 (1881 till 1904 by France, and then construction was taken over by the USA), the Panama Canal is a 77.1 km (48 mi) long **artificial ship canal (Cam 11,13)** that connects the Atlantic Ocean to the Pacific Ocean through Panama, thus creating a significantly important **route (Cam 12,13)** in international **maritime (Cam 13)** trade that cuts shipping time (and cost). The **canal** has a lock at each of its ends that lift ships to an artificial lake called Lake Gatun, which is 26 metres (85 ft) above sea level, and used to **traverse (Cam 11-traversed, Cam 10- traversing)** the land to the other ocean. When a ship reaches the other side, the second lock lowers it back to sea level

#### 4. The Great Pyramid of Giza

The Great Pyramid Of Giza is the oldest and largest between the 3 pyramids of Giza. It is the oldest and the only still standing structure of the Seven Wonders of the Ancient World, and inspired architects and engineers for thousands of years. The Great Pyramid was built around 2560 – 2540 BC, with a height of 146.5 metres (481 ft) and is listed as an UNESCO cultural world **heritage site (Cam 9)**. The **immense (Cam 9,11)** structure was constructed entirely using manual labor and for many centuries has been a source of great debates over the exact ways they were constructed: how the **enormous (Cam 11,13)** blocks were carried for long distances and to great heights without modern **machinery (Cam 9,10)**, how the **incredible (Cam 11)** precision was achieved, and more

#### 5. The Channel Tunnel (The Chunnel)

Opened in 1994, the Channel Tunnel is an undersea connection between England and France. Crossing the English Channel, the tunnel holds the record for longest undersea portion, at 37.9 km

**mankind** = all people, considered as a group

**artificial** = Artificial= synthetic, non-natural, man-made

**canal** = a long stretch of water made by people for boats to travel along

**route** = road =path = way

**maritime** = marine = relating to the sea

**Traverse**= pass through= to move across, over, or through something, especially an area of land or water

**Heritage**= the traditional beliefs, values, customs etc of a family, country, or society.

**Immense** = great, huge, enormous...

**Enormous** = **Immense** = great = huge

**machinery** = machines, often large machines

**incredible** = amazing = extraordinary

(23.5 mi) out of a total length of 50.5 km (31.4 mi). The tunnel reaches a depth of impressive 75 m (250 ft) and allow trains to travel at 160 kmh (99 mph). With the channel tunnel passengers can travel from London to Paris in just 2 hours 15 minutes!

## 6. The Great Wall Of China

The Great Wall Of China is the longest **man made (Cam 11)** structure, stretching for 8,850 kilometers. It was built in the 14th century and protected the northern borders of the Chinese Empire from the attacks of nomadic tribes. The Great Wall is sometimes falsely claimed to be seen from space with the naked eye. This construction is a **remarkable (Cam 9,12)** achievement that shows the **persistence (Cam 9)** and **capabilities (Cam 12)** of the Chinese Empire. The Great wall is the most famous wall and most famous man-made barrier in the world

## 7. Hoover Dam

Built in the years 1931 till 1936, during the Great Depression, the Hoover dam is one **spectacular (Cam 10,12) concrete (cam 12)** arch dam. It may not be one of the largest dams in the world today, but its construction in the middle of the desert in Nevada is remarkable and shows **determination (Cam 8)** and self belief. Its impact on it's surrounding, giving life to the once empty desert by means of electricity and **irrigation (Cam 10)** water, is priceless. The dam provides electricity to Nevada, Arizona, and California

## 8. Burj Khalifa Building

As of the beginning of 2015, the Burj Khalifa building in Dubai is by far the tallest freestanding structure in all of the world at a height of 828 meters (2,717 ft) and 163 floors. Completed in 2010, the **immense (Cam 9,11)** building is an engineering **accomplishment (Cam 10,11)** like no other,

**Manmade= artificial,**  
synthetic, manufactured #  
natural

**Remarkable=** extraordinary,  
amazing, outstanding,  
significant, incredible  
# ordinary

**Persistence=** determination to  
do something even though it is  
difficult or other people oppose  
it.

**Capabilities =** capacities =  
abilities = potentials

**Spectacular=** stunning,  
fantastic, impressive

**concrete =** a very hard building  
material made by mixing  
together cement, sand, small  
stones, and water

**determination =** the ability to  
continue trying to do  
something, although it is very  
difficult

**Irrigate=** to supply land or  
crops with water.

**Immense =** great, huge,  
enormous

designed to stand earthquakes, strong winds and other hazards. A building like that would seem completely impossible just a century ago

### 9. The Millau Viaduct

Bridge building is a key mean in overcoming natural obstacles like rivers and canyons. Overtime the technology of bridge building improved and as of 2015, the Millau Viaduct is at the front of that technology. The Millau Viaduct bridge is 343 metres (1,125 ft) tall, the tallest bridge in the world. Opened in 2004, this 4 lanes bridge is a cable-stayed bridge that spans the valley of the River Tarn near Millau in southern France. The bridge is the tallest structure in France, taller than the Eiffel Tower!

### 10. Delta Works flood barrier and the

#### Oosterscheldekering

The Delta Works is the largest flood **barrier (Cam 9)** project in the world and one of the largest **man-made** barriers in history, protecting a large land area in the Netherlands. It consists of number of means including dams and surge barriers. The Oosterscheldekering (Eastern Scheldt storm surge) is one of those surge barriers, and at a length of 9 km (5.6 miles) it is the largest surge barrier in the world. The Oosterscheldekering is located 1.5 hours drive from Amsterdam. The Delta Works is **crucial (Cam 9,11)** to the Netherlands as over a quarter of the country including over half its population is below sea level.

**Accomplishment** = achievement, success

**Barrier** = Block = Obstacle

**Crucial**= vital, very important, central...

## PASSAGE 3: REDUCING THE EFFECTS OF CLIMATE CHANGE

### ARTICLE 3: Geoengineering may be used to combat global warming, experts say

The world may increasingly look to **geoengineering** (Cam 11) in the wake of the latest UN **climate** (Cam 12,13) report, which says it could be **adopted** (Cam 12,13) as a **temporary** (Cam 13- temporarily) “remedial measure” if the world heads towards dangerous levels of warming.

The authors of the new 1.5C study by the Intergovernmental Panel on Climate Change say there is high agreement that the **injection** (Cam 10,11- inject) of millions of tonnes of **sulphur dioxide** (Cam 11) into the **stratosphere** (Cam 11) could help limit temperature rises to the most **ambitious** (Cam 11,12) target of the Paris accord.

But the authors warn there are major uncertainties about the social, environmental and **ecological** (Cam 10,11) impacts, which mean the world would be far better off if **policymakers** (Cam 13) strengthened natural cooling systems such as forest cover and **accelerated** (Cam 11- acceleration) efforts to reduce **carbon emissions** (Cam 11).

The lengthy document – which was approved at the weekend by all 195 nations in the UN – mentions several options for **man-made** (Cam 11) **interference** (Cam 9,12) in climate systems, including ocean fertilisation, carbon dioxide removal, **marine** (Cam 9,11) cloud brightening, cirrus cloud thinning and ground-based albedo modification.

**Geoengineering** = the study and activity of finding ways to change the earth's atmosphere in order to reduce global warming  
**Climate** = weather  
**Temporary** = Short-term = Momentary

**Injection** = Insertion = Addition  
**sulphur dioxide** = a poisonous gas that is a cause of air pollution in industrial areas

**Stratosphere** = a very high position

**Ambitious** = Determined = Striving = Motivated

**ecological** = relating to ecology or the environment

**policymakers** = a member of a government department, legislature, or other organization who is responsible for making new rules, laws

**Accelerate** = quicken, speed up # slow down

**carbon emissions** = carbon dioxide that planes, cars, factories, etc. produce, thought to be harmful to the environment

**man-made** = synthetic, Artificial # natural

**Interference** = an act of interfering (to deliberately get involved in a situation where you are not wanted or needed).

**Marine** = sea, oceanic, aquatic

**Stratosphere** = a very high position

But it focused most on **stratospheric (Cam 11) aerosol (Cam 11)** injection, a technique that essentially **mimics (Cam 11,13)** the effect of a volcano by pumping gas into the sky that turns into aerosols, which **reflect (Cam 13- reflection, Cam 12- reflecting)** part of the sun's heat.

Although the authors do not include such strategies in their pathways to 1.5C above **pre-industrial (Cam 8,11)** temperatures, they raise the possibility that it could be used as a **supplementary (Cam 8)** measure if this target is missed.

"If **mitigation (Cam 12- mitigate, mitigating)** efforts do not keep global mean temperature below 1.5C, solar **radiation (Cam 9,11)** modification can **potentially (Cam 9,12)** reduce the climate impacts of a temporary temperature overshoot, in particular **extreme (Cam 12,13)** temperatures, rate of sea level rise and **intensity (Cam 9,10)** of tropical cyclones, alongside intense mitigation and adaptation efforts," the report observes.

A search for palliatives will be necessary as the world is almost certain to miss the 1.5C goal. Current national pledges are forecast to lead to at least 3C of warming by the end of the century – and that is if governments keep their **commitments. (Cam 10,13)**

The IPCC is clearly hesitant to endorse such emergency measures in part because this could allow government leaders to continue pushing problems into the future, but also because of the **immense (Cam 11)** risks involved.

The report notes that the injection of sulphur dioxide would change rainfall patterns and weather **circulation (Cam 8- circulate)** as well as **disrupting (Cam 10- disruption)** **stratospheric** chemistry and ice formation. It could also result in more ultraviolet light

**Aerosol**= a metal container in which liquids are kept under pressure and forced out in a spray

**Mimic**= imitate, copy...

**Reflect** = throw back (heat, light, or sound) without absorbing it

**pre-industrial**= of or relating to a time before industrialization.

**Supplement**= add to, enhance, increase, make bigger

**Mitigate**= lessen, reduce, alleviate...

**Radiations** = a form of energy that comes especially from nuclear reactions, which is very harmful to living things

**Potential**= possible, likely

**Extreme**= severe, intense, excessive, harsh

**Intensity**= strength, power, amount...

**Commitment**= responsibility, promise

**Immense**=extremely large, enormous

**Circulation** = movement = flow

**Disrupt**= to prevent something, from continuing as usual

**Stratosphere** = a very high position

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| <p><b>exposure (Cam 11,13)</b>, which would have a negative impact on human health.</p> <p>Ethical and institutional questions also arise over who would oversee such operations and which areas would be affected. The report suggests a number of UN organisations as possible supervisory bodies. But authors also <b>observe (Cam 10,13 – observed)</b> that there are <b>scarcely (Cam 8- scarcely, Cam 12)</b> any laws or <b>regulations (Cam 8,10)</b> to stop any country that wants to push ahead by itself. The only guideline cited was the Convention on Biodiversity which states “no climate-related geoengineering activity that affects <b>biodiversity (Cam 12)</b> may take place.”</p> <p>There are doubts also over effectiveness. While the aerosols might constrain temperature rises, they would not stop the <b>accumulation (Cam 10,13- accumulate)</b> of <b>carbon dioxide</b> in the <b>atmosphere (Cam 11,13)</b> and the acidification of the oceans. What happens when this “temporary measure” is <b>halted (Cam 10,12 - halt)</b> is also an area of concern, as the planetary system might suddenly be hit by a <b>surge (Cam 8,12)</b> in temperature.</p> <p>The IPCC says these uncertainties constrain the ability to <b>implement (Cam 11- implemented)</b> solar radiation management in the near future. But with the 1.5C target current on course to be overshoot at some point between 2030 and 2052, the urgency is likely to grow.</p> <p>Johan Rockström, coauthor of the recent Hothouse Earth study, said the IPCC report was likely to <b>stimulate (Cam 11,12 – stimulated)</b> discussion of these extreme emergency measures.</p> <p>“I think this will raise solar radiation management to the highest political level. We currently have no <b>framework (Cam 11,13)</b> for this,” he said. “I’m</p> | <p><b>Exposure=</b> the fact of experiencing something or being affected by it</p> <p><b>Observe=</b> watch, view, scrutinize...</p> <p><b>Scarce=</b> rarely = hardly = barely</p> <p><b>Regulation =</b> rule = guideline = principle</p> <p><b>Biodiversity=</b> the variety of plants and animals in a particular place.</p> <p><b>Accumulate=</b> to gather, collect...</p> <p><b>Atmosphere=</b> air in environment</p> <p><b>Halt =</b> prevent= stop= discontinue</p> <p><b>Surge=</b> to increase suddenly.</p> <p><b>Implement=</b> put into practice, apply</p> <p><b>Stimulate=</b> accelerate, arouse, inspire, promote</p> <p><b>framework =</b> a system of rules, ideas, or beliefs that is used to plan or decide something</p> |
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| <p>very scared of this technology but we need to turn every stone now.”</p> <p>James Hansen said the tipping point in public opinion was more likely to come at a slightly higher temperature, but by then it may already be too late.</p> <p>“2C would force geoengineering on today’s young people. Geoengineering, if global temperature passes 2C, would start, at the latest, once ice sheet <b>collapse (Cam 11,12 – collapsed)</b> begins,” he told the Guardian.</p> <p>“Unfortunately, because of the inertia of the system, geoengineering then would probably be too late to <b>prevent (Cam 12,13)</b> locking in the eventual loss of coastal cities.”</p> | <p><b>Collapse</b>= breakdown, failure, end...</p> <p><b>Prevent</b>= stop.</p> |
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## TEST 2:

### PASSAGE 1: RAISING THE MARY ROSE

#### ARTICLE 1: Intrepid French hunt for sunken warships Cordelière and Regent

The **spirit** (Cam 8) of famed French explorer Jacques Cousteau lives on in a mission to find the **wrecks** (Cam 11) of two **warships** (Cam 9,11), one French and one English, that **sank** (Cam 11) 500 years ago off the Brittany port of Brest.

Instead of Cousteau's old minesweeper Calypso, it is the French culture ministry's surveillance ship André Malraux and its doughty crew of scientists and divers.

Today's adventure: to locate, **excavate** (Cam 11- **excavation**) and eventually raise the wrecks of the Cordelière and the Regent - two behemoths of the Tudor seas that sank together in the **Battle** (Cam 11,13) of Saint-Mathieu in 1512.

And filling the Cousteau role is Michel L'Hour, marine **archaeologist** (Cam 12,13) extraordinaire and veteran of a thousand missions to explore France's underwater **heritage** (Cam 9,13).

"I have been obsessed with finding these ships for 40 years," he says, ruddy-faced and bearded like any proper sea-dog.

**Spirit** = soul = essence

**Wreck**= a ship that has sunk

**warships** = a ship supplied with guns, for use in war

**sank**(past simple of sink) = to go down below the surface of water

**excavate** =dig

**Battle** = Combat = Conflict = Fight

**Archaeologist**= the person studying about ancient societies by examining what remains of their buildings, graves, tools etc

**Heritage**= the history, traditions and qualities that a country or society has had for many years and that are considered an important part of its character

"I am not so young any more, and I think this may be my last mission. If I can locate the ships, then leave them to my **colleagues (Cam 12,13)** to **excavate**, I will be a happy man."

## How the two ships were sunk

For the French, or rather for the people of Brittany, the Cordelière has mythic status. She was the flagship of the duchy's last **independent (Cam 12,13)** ruler and revered heroine, the Duchess Anne.

And she was captained up until the moment of sinking (and his death) by another Breton hero, Hervé de Portzmoguer, a kind of patriot-corsair. His Frenchified name Primauguet is still given to **vessels (Cam 11,13)** of the French navy to this day.

But the English have a hand in this tale too.

The Regent was, in its day, every bit as important as its sister ship the Mary Rose, which was famously raised from the Solent 36 years ago and is now **on display (Cam 11,13)** in Portsmouth.

If anything, the Regent was the bigger ship. And if Henry VIII's Mary Rose is anything to go by, then this would be a stupendous find indeed.

The trouble is no-one knows exactly where the Battle of Saint-Mathieu took place.

It was during one of the lesser-known wars between England and an alliance of France and a still-independent Brittany.

On 10 August 1512, the Franco-Breton **fleet (Cam 8,11)** was at anchor off Brest when it sighted the approaching English. Most of the French ships made haste to get to safety through the mouth of the inner bay of Brest, a passage known as the goulet.

**colleague** = coworker, partner, Collaborator

**independent** = self-reliant = self-sufficient

**vessel**= boat, sailing boat, ship, craft

**on display**= on show= something that is on display is in a public place where people can look at it

**Fleet**= ship in a navy

But the Cordelière and two other ships stayed to fight off the attack. The Regent bore down on the Cordelière and for two or three hours there was close-quarters fighting. But then, and no-one knows why, it all ended with a **massive (Cam 10,11) explosion (Cam 9,11)**.

The two ships, **entangled (Cam 13- entangling, Cam 8- entanglement)** in battle, sank together to the bottom. Hundreds died.

### Out there, somewhere

With backing from the French government, the Brittany region, universities and industry, Michel L'Hour has **assembled (Cam 11)** a multidisciplinary team to find what he knows must be down there somewhere.

**Tides (Cam 9,10) and currents (Cam 9,11)** have been recalculated. Previous efforts had failed to spot that the calendar changed in 1582.

Naval charts of the sea floor have been **re-examined (Cam 10,13- examined)**. Historians are looking through the few **contemporary (Cam 9)** accounts of the battle, and the search is on for clues from parish and other archives.

**Massive**= huge, enormous, very big...

**Explosion**= bang

**entangled** = involved with something or someone in a way that makes it difficult to escape

**Assemble**= bring together, put together, gather

**Tide**= wave.

**Currents** = flow = stream

**re-examined** = to look at something in detail again

**Contemporary**= someone who lived or was in a particular place at the same time as someone else.

## PASSAGE 2: WHAT DESTROYED THE CIVILISATION OF EASTER ISLAND

### ARTICLE 2: New theory paints more sophisticated picture of ancient Easter Island

For decades, **mystery** (Cam 12,13) has swirled around what happened to the founding population of the remote Easter Island, known for its towering stone statues **depicting** (Cam 12) large carved heads. A long-held theory suggests that after the islanders **set up** (Cam 9,13) camp and **carved** (Cam 10,11) the giant statues, they destroyed their own society through infighting and a **depletion** (Cam 11-depleted) of natural resources.

But a new study suggests a different **scenario** (Cam 11) entirely, and the islanders could have a more **complex** (Cam 12,13) history than previously believed.

Many aspects of the island, the statues and the Polynesian **seafarers** (Cam 11) who arrived there 900 years ago have been studied over the years. But a study published Monday in the Journal of Pacific Archaeology uses multiple data sets from recent **excavations** (Cam 11) to provide a better understanding of the society that created the **statues** (Cam 11) and how they were carved.

So the researchers used an unlikely method to study the society and discern what might have happened: a chemical analysis of stone tools used on the statues.

What they learned from **excavating** four of the statues and the volcanic stone basalt tools used to **carve** them painted a different picture: a **sophisticated** (Cam 11- sophistication, Cam 13- sophisticated) and **collaborative** (Cam 9,10) society.

**Mystery**= unknown, unidentified...

**Depict**= show, represent, describe, illustrate...

**Set up**= start, establish

**Carve**= to make something by cutting into especially wood or stone, or to cut into the surface of stone, wood, etc.

**Scenario**= the description of possible actions or events in the future

**complex** = complicated, intricate

**seafarers** = a person who travels by sea

**Excavation**= digging

**Statue** = Sculpture.

**Sophisticated**= advanced, highly developed, **complex**, complicated, elaborate

**Collaborative**= joint, two-away, shared...

"The idea of competition and **collapse (Cam 8,11)** on Easter Island might be overstated," said lead study author Dale Simpson Jr., an **archaeologist (Cam 12,13)** at the University of Queensland, in a statement. "To me, the stone carving industry is **solid evidence (Cam 13)** that there was **cooperation (Cam 13)** among families and **craft (Cam 13)** groups."

A **complex** society

Easter Island is 2,300 miles off the coast of Chile. About 900 years ago, the founding population landed on the island, called Rapa Nui in the local language. These Polynesian seafarers came on two canoes and were led by Hotu Matu'a, who would become the island's first **chief (Cam 11)**, according to oral tradition.

The population soared to include thousands of people, and they carved full-body figures, called moai, to **represent (Cam 13- representation, Cam 12- represented)** important Rapa Nui **ancestors (Cam 11)**. There are nearly a thousand statues, many buried up to their heads due to the passage of time. The largest statue is over 70 feet tall.

Their **sheer (Cam 10)** size and number is **indicative (Cam 9,13)** of a complex, sophisticated society, according to the researchers.

"Ancient Rapa Nui had chiefs, priests, and guilds of workers who fished, farmed, and made the moai. There was a certain level of sociopolitical organization that was needed to carve almost a thousand statues," Simpson said.

During the statue excavations, Jo Anne Van Tilburg, director of the Easter Island Statue Project, and her Rapa Nui excavation team recovered about 1,600 stone tools. They did a close chemical and mass spectrometer analysis of **fragments (Cam 9, Cam 10- fragmentation)** from 17 of the tools, called toki.

"We wanted to **figure out (Cam 12, 13)** where the **raw materials (Cam 11,12)** used to manufacture the artifacts came from," said Laure Dussubieux, a study co-author and Field

**Collapse**= breakdown, failure, end

**Archaeologist**= the person studying about ancient societies by examining what remains of their buildings, graves, tools etc.

**Solid**= strong, certain, firm, concrete

**evidence** = anything that helps to prove that something is or is not true

**Cooperation**= collaboration

**Craft**= a job or activity in which you make things with your hands, and that you usually need skill to do

**chief** = the person in charge of a group or organization

**Represent**= stand for, symbolize

**Ancestor**= a person related to you who lived a long time ago

**Sheer** = absolute = complete

**Indicative**= a sign that something exists, is true, or is likely to happen

**Fragment**= a small piece of something that has broken off or that comes from something larger.

**Figure out** = to understand or solve something

**Raw**=fresh = unprocessed, unrefined, untreated...

Museum scientist, in a statement. "We wanted to know if people were taking material from close to where they lived."

The location is key because there are three sources, or quarries, where the islanders could have **gathered (Cam 12, Cam 13 -gathering)** material for tools. The basalt quarries cover the size of two football fields. Basalt samples reveal their sources through their chemical **elements (Cam 12,13)**, which link back to the geology of the site.

"The majority of the toki came from one quarry complex -- once the people found the quarry they liked, they stayed with it," Simpson said. "For everyone to be using one type of stone, I believe they had to **collaborate (Cam 9,10)**. That's why they were so successful -- they were working together."

Looking ahead

This contrasts with the idea that the residents ran out of resources and fought amongst themselves, the researchers said. But later on, it is believed that the arrival of **colonists (Cam 12- colonised)** and the **institution (Cam 10,12)** of slavery **decimated (Cam 11)** the population.

"There's so much **mystery** around Easter Island, because it's so **isolated (Cam 11)**, but on the island, people were, and still are, **interacting (Cam 8,13)** in huge amounts," Simpson said.

"There are thousands of Rapa Nui people alive today -- the society isn't gone."

But the researchers also urge caution and believe that this study is just the beginning, laying the groundwork for more research.

"The near **exclusive (Cam 11,13)** use of one quarry to produce these seventeen tools supports a view of craft specialization based on information exchange, but we can't know at this stage if the interaction was collaborative," Van Tilburg said. "It may also have been coercive in some way. Human behavior is complex. This study encourages further mapping and stone sourcing, and our excavations continue to **shed new light on (Cam 11)** moai carving."

**Gather** = Accumulate, collect

**Element**= factor.

**collaborate** = cooperate = work in partnership = work together

**colonists** = someone who settles in a new colony

**Institution**= a large organization that has a particular kind of work or purpose.

**Decimate**= destroy, devastate, ruin

**Isolated**= remote

**Interacting** = relating= networking = cooperating

**Exclusive**= available or belonging only to particular people, and not shared.

**shed light on** = explain = make clear = clarify =

## PASSAGE 3: NEUROAETHETICS

### ARTICLE 3: Neuroaesthetics

Willem de Kooning, the famous Dutch-American **abstract** (Cam 11.12) expressionist painter, continued to paint for several years after developing Alzheimer's disease. But his paintings were different somehow, more deeply expressing his abstract style, according to some art critics. De Kooning's change in style **explicitly** (Cam 8, 12 – **explicit**) makes the case that art can serve as a window into the workings of the human brain, that when the brain changes, so too do artistic expression and perception.

But what in the brain **triggers** (Cam 12,13) **aesthetic** (Cam 10- **aesthetically**, Cam 11) experiences? And how does knowledge of basic brain **mechanisms** (Cam 11,12) **inform** (Cam 8,13) our understanding of these experiences? These questions are at the heart of an **emerging discipline** (Cam 11) **dedicated to** (Cam 10,12) exploring the neural processes underlying our appreciation and production of beautiful objects and artwork, experiences that include **perception** (Cam 9,12), **interpretation** (Cam 9,11), emotion, and action. This new field represents a convergence of neuroscience and empirical aesthetics—the study of aesthetics rooted in observation—and is dubbed neuroaesthetics, a term **coined** (Cam 11) in the 1990s by vision **neuroscientist** (Cam 9,12) Semir Zeki of University College London.

Neuroaesthetics is both **descriptive** (Cam 8,9) and experimental, with **qualitative** (Cam 10) observations and quantitative tests of **hypotheses** (Cam 13) , aimed at advancing

**Abstract**= non-realistic, symbolic

**Explicitly** = clearly = obviously

**Trigger**= activate, generate, cause...

**Aesthetic** = relating to the enjoyment or study of beauty

**Mechanism** = process, system, operation

**Inform** = tell = notify

**Emerging** = in an early state of development

**Discipline** = regulation = control = strictness

**Dedicate**= to give all your attention and effort to one particular thing.

**Perception**= view, opinion...

**Interpretation**= explanation, understanding

**Coined** = invented

**neuroscientist** = a scientist who studies the nervous system and the brain

**Descriptive**= describing how the words of a language are actually used, rather than saying how they ought to be used= expressive= illustrative

our understanding of how humans process beauty and art. While the field is still young, interest is growing, as evidenced by several recent books on the topic. Moreover, recent workshops such as “Pain and Pleasure: The Art and Science of Body Representation,” held in Venice last November, and “Visual Arts and the Brain,” held at the Royal Society of Medicine in London the following month, **demonstrate (Cam 9,13)** the international scope of this discipline as it **addresses (Cam 12,13)** various aesthetic experiences, and their underlying neural correlates, in health and in disease.

### Art and the brain

Early **neuroaesthetic (Cam 11)** writings in the late 1990s by Zeki and University of California, San Diego, neuroscientist Vilayanur Ramachandran **identified (Cam 12,13)** parallels between an artist’s **approach (Cam 12,13)** to her visual world and her brain’s processing of visual information. Light entering our eyes is segregated into a number of elemental **properties (Cam 10,13)**, such as luminance, color, and motion, that are processed in different visual centers in our brains. At the turn of the 20th century, artists played with these elements in their artwork. For example, the French artists Henri Matisse and André Derain highlighted color to express emotions, and the American sculptor Alexander Calder **isolated (Cam 11)** visual motion in his mobiles (below), whose **suspended (Cam 13) components (Cam 11,13)** were delicately positioned to be moved by air currents or motors.

Artists often **depict (Cam 13-depicted, depiction)** mental **representations (Cam 9)** of an object rather than its physical form. Their renditions do not **adhere (Cam 9- adherents)** strictly to the light, shadow, and color properties of objects in the physical world, yet they **appeal (Cam 11- appealing)** to us. Shadow contours are too **fleeting** and changeable to provide

**qualitative** = relating to the quality or standard of something rather than the quantity

**hypotheses** = theories = assumptions

**Demonstrate**= show, indicate, prove

**Address**= solve, tackle, deal with

**Neuroaesthetics**= a field of study in which researchers attempt to understand how the brain responds to art

**Identify**= discover, find, detect...

**approach**= move toward, come close to

**Property**= a quality or characteristic that something has

**Isolated**= remote

**Suspended** = postponed

**Component**= part, factor, element

**Depict**= show, represent, describe, illustrate...

**Representation**= picture, painting, drawing, picture, illustration

**Adherent**= supporter= believer= devotee

**Appeal** = attract = fascinate = interest = charm

**Fleet**= ship in a navy

**Reliable**= dependable, trustworthy, good

#unreliable, untrustworthy

**reliable (Cam 10,12)** information about real-world objects, so our brains never **evolved (Cam 11,13)** to be **sensitive (Cam 11,13)** to the shape of shadows. As a result, inaccurately shaped shadows in works of art are not displeasing to our eyes. By contrast, artists are typically careful to **depict** shadows as having less luminance than the object casting the shadow, as people are sensitive to the brightness of objects relative to shadows.

Moreover, some artists, at least implicitly, use perceptual “tricks.” One such trick is the peak-shift principle, in which artists **exaggerate (Cam 13- exaggerated)** certain features through the use of shape, shading, or color. The idea that exaggerated features **enhance (Cam 13- enhanced)** an intended response was first **articulated (Cam 11- articulate)** in 1954 by the ethologist Nikolaas Tinbergen, who observed that seagull chicks more **vigorously (Cam 10- vigorous)** peck a disembodied long, thin stick with three red stripes at the end than they do at their mother’s beak, which has a red spot near the tip. But the use of the peak-shift principle in art predates Tinbergen. This perceptual mechanism can be seen in bronze **sculptures (Cam 10)** of the 12th-century Chola **dynasty (Cam 11)** in India, where the goddesses’ large breasts and hips and narrow waists enhance female sensuality, grace, poise, and dignity.

Artists also capitalize on the way the human visual system processes information in two interacting neural systems referred to as the “what” and the “where” streams. Form and color are processed in the “what” stream, revealing an object’s identity. Luminance and motion are processed in the “where” stream, which **reveals (Cam 12,13)** an object’s location. Viewers of some **impressionist paintings (Cam 11)**, such as Claude Monet’s *Impression, Sunrise* (opposite page), observe that the effect of shimmering on water surfaces or the sun’s glow on the horizon appears because the

**Evolve**= develop, change, grow...  
**Sensitive**= easily offended, easily upset, easily hurt

**Depict**= show, represent, describe, illustrate...

**Exaggerated**= being described as better, larger etc than it really is

**Enhance** = increase = boost = strengthen = enrich

**Articulate**= speak out.

**Vigorous**= strong and healthy...

**Sculpture**= statue.

**dynasty** = a series of rulers or leaders who are all from the same family, or a period when a country is ruled by them

**Reveal**= bring to light, show, point out

**impressionist paintings** = a style of painting, which began in France in the 1860s, in which the artist tries to represent the effects of light on an object, person, area of countryside, etc

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| <p>objects are painted with the same luminance but different colors. While the brain can identify these objects, their location is hard to fix, because the “where” stream does not register objects that have the same brightness as their background. Consequently the water and the sun in the Monet painting are seen as unstable in their location, lending the painting that shimmering quality.</p> |  |
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## TEST 3

### PASSAGE 1: THE STORY OF SILK

#### ARTICLE: Debate (cam 9,11): What China's new Silk Road means for Europe

China's Belt and Road Initiative (BRI) is the biggest **infrastructure** (cam 12,13) project in human history. This overarching program – which includes overland **routes** through Central Asia in the Silk Road Economic Belt, plus a seaborne **component** (cam 8,11,12,13) called the Maritime Silk Road that **skirts** (cam 12) the southern rim of Asia on the Indian Ocean route to the Middle East and Europe – is central to China's **ambition** (cam 10) of becoming the preeminent world power.

Its **completion** (cam 9), planned for the 100<sup>th</sup> anniversary of the founding of the Chinese People's Republic in 2049, will **cement** (cam 11) Beijing's influence over the huge Eurasian **continent** (sub-continent – cam 13) and parts of Africa, containing some 65 countries and more than 4 billion people.

The terminus of the BRI's overland and **maritime** (cam 13) **segments** (cam 10,11) is Europe, which has had “**virtually** zero” influence on the **mammoth** (Cam 8,10) project. The passivity of one of the world's biggest economic, political and strategic players mirrors China's own passivity when challenged by the West in the 18<sup>th</sup> and 19<sup>th</sup> centuries, as Prince Michael of Liechtenstein, the founder and chairman of Geopolitical Information Services (GIS), noted in a recent analysis.

To discuss the new Silk Road's **implications** (cam 8) of this project for Europe, GIS brought together leaders from Polish business and

**Debate**= argument, discussion...

**Infrastructure**= the basic systems and structures that a country or organization needs in order to work properly, for example roads, railways, banks etc.

**route** = road =path = way

**Component**= part, factor, element

**skirts** = to go around or move along the edge of something

**Ambition**= aim, goal, objective...

**Completion** = achievement = accomplishment

**Cement** = strengthen = boost

**Continent** = area = region = zone

**maritime** = marine = relating to the sea

**Segment**= section, part, piece, sector

**Virtually**= almost, nearly

**Mammoth**= enormous, huge, massive...

**Implications** = consequences = significance

politics for a debate at the Warsaw headquarters of the daily newspaper *Rzeczpospolita* on Nov. 20, 2017. The **distinguished** (cam 10) **panel** (cam 8) of speakers included:

- Ryszard Czarnecki, vice president of the European Parliament
- Grzegorz Kurdziel, management board member of Poczta Polska SA
- Cornelius Ochmann, director of the Foundation for Polish-German Cooperation
- Radosław Pyffel, Polish representative and **alternate** (cam 12) director at the Asian Infrastructure Investment Bank (AIIB)
- Karol Zarajczyk, CEO of Ursus SA, a Warsaw-based **tractor** (cam 11) and agricultural machines manufacturer

“Europe is seeing a **reversal** (reverse – cam 10,11) in the economic pecking order, with investment and influence spreading from east to west

While these points were **acknowledged** (acknowledging –cam 11) by Mr. Pyffel, Poland’s **representative** (Cam 13- representation, Cam 12- represented) on the AIIB, he noted that Chinese plans and the Western response had already gone “beyond the discussion stage.” At this point, China’s **expansion** (cam 10,12) is **irreversible** (cam 10) – the main question is “whether we take part or not.” He noted that when China found itself in an analogous position in the 1990s, it was careful to protect its **internal** (cam 8,9) market. Europe would also do well to take a pro-active role in developing business **strategies** (cam 12,13) to get the most benefit from BRI, as Poczta Polska was now doing. “Otherwise, when the next BRI summit is held in 2019, Chinese bureaucrats will be the ones coming up with ideas,” he said

**Distinguished**= differentiate, tell apart...

**Panel** = a small group = team = board = committee

**Alternate**= substitute, different...

**Tractor**= a strong vehicle with large wheels, used for pulling farm machinery

**Reversal** = turn around

**Acknowledged** = admitted = recognized

**representative** = someone who speaks or does something officially for group of people

**Expansion**= growth, development, increase, spreading out...

**Irreversible**= irreversible damage, change etc is so serious or so great that you cannot change something back to how it was before.

**Internal**= inside, inner...

**Strategy**= plan.

## PASSAGE 2: GREAT MIGRATIONS

### ARTICLE: Why migratory birds?

Avian migration is a natural miracle. Migratory birds fly hundreds and thousands of kilometres to find the best **ecological** (cam 10,11) conditions and **habitats** (cam 11) for feeding, **breeding** (cam 12) and raising their young. When conditions at breeding sites become unfavourable, it is time to fly to regions where conditions are better.

There are many different migration **patterns** (cam 12). The **majority** (cam 11) of birds migrate from northern breeding areas to southern wintering grounds. However, some birds breed in southern parts of Africa and migrate to northern wintering grounds, or **horizontally** (cam 10), to enjoy the **milder** (mild-cam 8, mildly –cam 12) **coastal** (cam 13) climates in winter. Other birds reside on lowlands during the winter months and move up a mountain for the summer.

Migratory birds have the perfect morphology and physiology to fly fast and across long distances. Often, their journey is an **exhausting** (exhausted – cam 13) one, during which they go to their limits. The Red Knot has one of the longest total migration **routes** (cam 8,13) of any bird, travelling up to 16,000 kilometres twice a year. It breeds in Siberia and overwinters on the west coast of Africa, some even going down to the tip of South Africa.

It is truly amazing how migratory birds can **navigate** (navigation –cam 8, navigator-cam 13) with pin-point **accuracy** (cam 11). Exactly how migrating birds find their flyways is not fully understood. It has been shown that they are able to orientate by the sun during the day, by the stars at night, and by the geomagnetic field at any time. Some species can even **detect**

**ecological** = relating to ecology or the environment

**Habitat**= the natural home of a plant or animal.

**Breed**= reproduce= if animals breed, they mate in order to have babies.

**Pattern**= the regular way in which something happens, develops, or is done.

**Majority** = main, popular, common

**horizontally** = parallel to the ground or to the bottom or top edge of something

**Milder** = slighter

**coastal** = in the sea or on the land near the coast (coast : where the land meets the sea)

**Exhaust**= run out, use up, deplete

**route** = road =path = way

**Navigate**= to understand or deal with something complicated.

**Accuracy**=exactness, precision...

**Detect** = recognize, discover, find

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| <p>(cam 9,13) polarized light, which many migrating birds may use for navigation at night.</p> <p>Migration is a perilous journey and involves a <b>wide range of (cam 8,12)</b> threats, often caused by human activities. And as <b>diverse (cam 9,10)</b> as people and their habits in different countries are, so are threats the birds face. As migratory birds depend on a range of sites along their <b>distribution (cam 11,12)</b> area, the loss of wintering and stopover sites could have a dramatic <b>impact on (cam 11,13)</b> the birds' chances of survival.</p> <p>Flying long distances involves crossing many borders between countries with differing environmental politics, <b>legislation (cam 9)</b> and conservation measures. It is <b>evident (cam 12,13)</b> that international cooperation among governments, NGOs and other stakeholders is required along the entire flyway of a species in order to share knowledge and to <b>coordinate (cam 9)</b> conservation efforts. The legal <b>framework (cam 11,13)</b> and coordinating instruments necessary for such <b>cooperation (cam 13)</b> is provided by multilateral environmental agreements such as CMS and AEWA.</p> <p>World Migratory Bird Day has a global outreach and is an effective tool to help raise global awareness of the threats faced by migratory birds, their ecological importance, and the need for international <b>cooperation to conserve (cam 13)</b> them.</p> | <p><b>a wide range of</b> = many, various, numerous...</p> <p><b>Diverse</b>= different, varied...</p> <p><b>Distribution</b>= spreading, allocation</p> <p><b>impact on</b> = influence = affect</p> <p><b>Legislation</b> = law= rule...</p> <p><b>evident</b>= clear= obvious= visible = <b>apparent</b></p> <p><b>Coordinate</b>= collaborate = to organize an activity so that the people involved in it work well together and achieve a good result.</p> <p><b>framework</b> = a system of rules, ideas, or beliefs that is used to plan or decide something</p> <p><b>Cooperation</b>= collaboration</p> <p><b>conserve</b>= <b>preserve</b>, protect, maintain, save, safeguard</p> |
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# **PASSAGE 3: PREFACE TO “HOW THE OTHER HALF THINKS: ADVENTURES IN MATHEMATICAL REASONING”**

## **ARTICLE 3: The first four standard**

### **Standard 4: Reasoning**

*All students will develop reasoning ability and will become self-reliant, independent mathematical thinkers.*

#### **Descriptive (Cam 8,9) Statement**

Mathematical reasoning is the **critical (cam 10,12)** skill that enables a student to make use of all other mathematical skills. With the development of mathematical reasoning, students recognize that mathematics makes sense and can be understood. They learn how to **evaluate (cam 9)** situations, select problem-solving **strategies (cam 12,13)**, draw logical conclusions, develop and describe solutions, and recognize how those solutions can be applied. Mathematical reasoners are able to **reflect on (cam 9,10)** solutions to problems and **determine (cam11,13)** whether or not they make sense. They **appreciate (cam 11)** the **pervasive (cam 10)** use and power of reasoning as a part of mathematics.

#### **Meaning and Importance**

There are various terms used to refer to "reasoning": critical thinking, higher-order thinking, logical reasoning, or simply reasoning. Different subject areas tend to use different terms. Across all of these subject areas, however, there are commonalities. The following phrases often appear in discussions of how reasoning is used (**adapted (cam 8,11)** from Resnick, 1987, pp 2-3):

- *Nonalgorithmic* - The **route (Cam 12,13)** to a solution is not fully **specified** in advance.

**Descriptive**= describing how the words of a language are actually used, rather than saying how they ought to be used= expressive= Illustrative

**Critical**= significant, important, vital, crucial, essential...

**Evaluate** : to judge how good, useful,or successful something is= assess= appraise

**Strategy**= problem, project, plan.

**reflect on** = to affect other people's opinion of someone or something

**Determine** = decide = specify

**Appreciate** = acknowledge, recognize, be aware of

**Pervasive**= existing everywhere.

**Adapted from** = created = designed

**route** = road =path = way

**specified** = detailed = identified

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| <ul style="list-style-type: none"> <li>• <b>Complex (Cam 12,13)</b> - The complete path to a solution is not fully <b>apparent (cam 8,9)</b> from any single vantage point.</li> <li>• <b>Multiple criteria (cam 10)</b>- The conditions established in the problem may <b>conflict (cam 11)</b> with one another.</li> <li>• <b>Uncertainty</b> - Not everything that <b>bears (cam 8)</b> on the task at hand is known.</li> <li>• <b>Imposing meaning</b> - The individual must find structure in <b>apparent</b> disorder.</li> <li>• <b>Effortful</b> - There is considerable mental work involved in the <b>elaborations (cam 8)</b> and judgments required.</li> <li>• <b>Self-regulation (regulation -cam 8,10)</b>- The individual monitors his or her own progress, and determines the appropriate course of action.</li> <li>• <b>Multiple solutions</b> - There is no single "best" solution; rather, there are many solutions, each with costs and benefits.</li> <li>• <b>Nuanced judgment</b> - The results must be <b>interpreted</b> (interpreting, interpretation, interpret – cam 9,10,11,12).</li> </ul> <p><b>Development and Emphases</b></p> <p>Every student has <b>potential (cam 12,13)</b> for higher-order thinking. The key is to unlock the world of mathematics through a student's natural <b>inclination (cam 11)</b> to strive for purpose and meaning. Reasoning is fundamental to the knowing and doing of mathematics. <b>Conjecturing (Conjectures – cam 9)</b> and demonstrating the logical validity of conjectures are the <b>essence (cam 10,11,13)</b> of the creative act of <i>doing</i> mathematics. To give more students access to mathematics as a powerful way of making sense of the world, it is essential that an <b>emphasis (cam 10,12)</b> on reasoning pervade all mathematical activity. In order to become confident, self-reliant mathematical thinkers,</p> | <p><b>complex</b> = complicated, intricate ...</p> <p><b>Apparent</b>= evident= clear= obvious= visible</p> <p><b>Criteria</b> = standards = principles ..</p> <p><b>Conflict</b>= contradict, disagree, differ</p> <p><b>Bears on sth</b> = to be connected or related to something</p> <p><b>Imposing</b> = large, impressive, and appearing important</p> <p><b>Elaboration</b>=intricacy, complication, complex...</p> <p><b>Self-regulation</b> = the act of making certain yourself that you act according to the rules, rather than having this done by other people</p> <p><b>Interpret</b>= understand, comprehend, make sense of, explain</p> <p><b>Potential</b>= possible, likely</p> <p><b>Inclination</b>= a feeling that makes you want to do something = tendency, proclivity</p> <p><b>Conjecture</b>= guess, hypothesis...</p> <p><b>Essence</b>= the most <u>basic</u> and important quality of something</p> <p><b>Emphasis</b>= stress = special attention or importance</p> |
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| <p>students need to develop the <b>capability (cam 12)</b> to confront a mathematical problem, <b>persevere (perseverance- cam 8)</b> in its solution, and evaluate and <b>justify (unjustified – cam 8)</b> their results.</p> <p>In summary, mathematical reasoning is the glue that <b>binds (binding –cam 12)</b> together all other mathematical skills. By using inductive and <b>deductive (deduction – cam 10)</b> reasoning as they learn mathematical concepts and solve (cam 12,13) mathematical problems, students come to recognize the <b>extent (cam 11,12)</b> to which reasoning applies to mathematics and to their world.</p> | <p><b>capability</b> = capacity, ability</p> <p><b>persevere</b> = persist = continue doing something in a determined way, despite having problems</p> <p><b>justify</b> = to show that something is reasonable, right, or true</p> <p><b>binds together</b> = unite = tie</p> <p><b>deductive</b> = logical = reasonable</p> <p><b>Extent</b>= degree, level</p> |
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## TEST 4:

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| <p><b>PASSAGE 1: RESEARCH USING TWINS</b></p> <p><b>ARTICLE 1:</b></p> <p>A new documentary <b>chronicles (cam 9)</b> the lives of triplets separated at birth in a <b>controversial (cam 10)</b> study — here's how scientists continue to use twins in research</p> <p>Twins have always fascinated scientists, especially as the subjects of studies about the influence of environment and genetics.</p> <p>The new documentary "Three Identical Strangers," which is now in theaters, tells the story of the <b>ultimate (cam 9,13)</b> test of nature versus <b>nurture (cam 8,11)</b>: it follows three <b>identical (cam 11,12)</b> brothers who were separated at birth and raised by different families.</p> <p>In 1980, two of the brothers met while attending Sullivan County Community College, and after making headlines, found the third <b>triplet (tripled –cam12)</b>. Aside from looking alike, the three shared similar behavioral quirks and preferences.</p> <p>But they were not the only twins in the study — according to NPR, of the 13 children <b>involved</b>, three sets of twins and one set of triplets have discovered one another. A book titled "Identical Strangers: A Memoir of Twins Separated and <b>Reunited (cam 11)</b>" was published in 2007 by a pair of twins who were also involved, Elyse Schein and Paula Bernstein.</p> <p>The other four subjects still do not know they have <b>identical twins (cam 11)</b>.</p> <p><b>A controversial twin study</b></p> | <p><b>Chronicle</b>= story, record...</p> <p><b>Controversial</b>= causing a lot of disagreement, because many people have strong opinions about the subject being discussed.</p> <p><b>Ultimate</b> = crucial = vital = decisive</p> <p><b>Nurture</b>= the education and care that you are given as a child, and the way it affects your later development and attitudes</p> <p><b>Identical</b> = exactly the same</p> <p><b>Triplet</b>= one of three children born to the same mother at the same time</p> <p><b>Involve</b>= include, contain, comprise...</p> <p><b>Reunite</b>= bring back together, come together</p> <p><b>Identical twin</b>= one of a pair of brothers or sisters born at the same time, who develop from the same egg and look almost exactly alike</p> |
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The study was conducted by child psychiatrist Peter Neubauer and Violet Bernard, a child psychologist. They worked with the Louise Wise Agency, which matched Jewish orphans with **adoptive (adoption –cam 12) families**, to **craft (cam 12,13)** a secret experiment that would test how much of a person's behavior is genetically influenced and how much is environmentally influenced.

The researchers carefully controlled which families the identical siblings **ended up (end up–cam9,13)** in, withheld information about their biological parents, and didn't tell the adoptive families that the children were twins or had siblings. Instead, they told the families that their children were being followed for a study about the development of adopted children.

The study ultimately ended in 1980, and because of the fear of backlash and **controversy** over **ethics (cam13)** and consent, Neubauer never published the results. The data is **sealed (cam 8,10)** in a Yale archive until 2066.

This is the only twin study that followed its subjects from **infancy** (infant –cam11,13), but it's far from the only time scientists have used genetically identical siblings in research.

Why scientists are obsessed with twins

The annual Twins Days Festival in Twinsburg, Ohio attracts the largest hoard of twins from all over the world. And you bet scientists are in attendance as well.

Conducted **properly (cam 10,11)**, twin experiments can give scientists **insights (cam 13,11)** into how different habits, treatments, or lifestyles affect two people with the same **genetic makeup (cam 11)**. Studying identical twins can also help scientists **pinpoint (cam 13)** the effect of **epigenetics (cam 11)**, or environmental influences, on gene expression

**Adoptive families:** families with a child or children who have been adopted (an adopted child has been legally made part of a family that he or she was not born into)

**Craft** = make = create = produce

**End up=** to be in a particular place or situation, state after a series of events, especially when you did not plan it.

**Ethics** = morals

**Seal=** shut out, close up, stop entering

**infancy** = the time when someone is a baby or a very young child

**Properly** = right = suitably = correctly

**Insight=** perception, awareness, understanding, comprehension

**Genetic makeup** = genetic combination

**Pinpoint=** locate, identify, find

**Epigenetics=** a branch of genetics that studies the chemical reactions that turn genes on and off

**Determine** = decide = specify

**traits** = character, personality, qualities...

and function. This can help **determine (cam 8,11)** if certain **traits (cam 10)** or diseases lean more heavily on genetics or the environment.

In history, twins have been used in research about I.Q., everyday diseases, eating disorders, **obesity (cam 13)**, developmental and psychological traits, and sexual orientation, according to Smithsonian Magazine.

In a **comprehensive (cam 8,13)** review of twin studies worldwide, which was published in Nature Genetics in 2015, researchers found that on average, environment and genetics have a 50/50 influence on a person's traits and disease. But certain conditions like bipolar disorder **rely (cam 13)** more heavily on genetics.

Today, twin studies are still commonly used. There are studies on mood and anxiety disorders as well as **asthma (cam 12)** and allergies. The Minnesota Center for Twin and Family Research collects community-contributed data from twins that helps them map out mental health outcomes and examine the development of **substance (cam 8)** use and related behavior disorders. For one study, they examined personality development of twins to see whether environment or genes **played a role in (cam 9)** risk-taking behaviors that lead to **substance** abuse.

Most famous, however, are NASA's twin studies. After astronaut Scott Kelly got back from a year in space, scientists **observed (cam 10,13)** that 7% of his genes were expressed differently than those of his **identical** twin. The genes that were **altered (cam 9)** were related to the **immune (cam 8)** system, bone formation, DNA repair, and responses to an oxygen-**depleted (cam 11)** or carbon-dioxide rich environment.

On top of this, Scott Kelly's telomeres — the caps at the ends of our chromosomes that affects cell aging — appeared to get longer in space, but they shrunk back when he returned

**Obesity**= fatness, overweightness

**Comprehensive**= thorough, in-depth, complete #partial, limited

**rely on**= depend on

**asthma** = a chest disease in which breathing can become difficult, often caused by an allergic reaction

**Substance** = element, material, ingredient...

**Play a role in**= to have an effect or influence on something = influence, impact, affect

**Observe**= watch, view, scrutinize...

**Altered**= changed # original

**Deplete**= reduce, eat up, lessen, exhaust, diminish # increase

|                                                                                                                                                                                                                                                                                                                                                       |                                                                                                           |
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| <p>to Earth. His gut also hosted different bacteria, and he returned two inches taller.</p> <p>NASA recently sent 20 mice into space while their twins stay on Earth. In partnership with astronauts on the International Space Station, agency scientists plan to study changes in the <b>rodents (cam 11)</b> microbiomes and circadian cycles.</p> | <p><b>rodents</b> = any of various small mammals with large, sharp front teeth, such as mice and rats</p> |
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## PASSAGE 2: AN INTRODUCTION TO FILM SOUND

### ARTICLE 3: Sound in Filmmaking

“Sound” **refers (cam 11)** to everything we hear in a movie — words, sound effects, and music. Sound is used in film to heighten a mood, provide us with information about the location of a **scene (cam 11)**, advance the **plot (cam 13)**, and tell us about the characters in the story.

There are two **categories (cam 9)** of sound in film: Diegetic and Non-Diegetic. Diegetic Sound refers to all those audio **elements (cam 8)** that come from sources inside the world we see on the screen, including dialogue, doors slamming, footsteps, etc. Non-Diegetic Sound refers to all those audio elements that come from outside of the **fictional (fiction –cam 11)** world we see on screen, including the musical score and sound effects like the screeches in the shower scene in *Psycho*.

How do Sound Effects help to Shape a Film?

Sound effects can be used to add mood or **atmosphere (cam 8)** to a film by creating a soundscape that accents or adds another layer of meaning to the images on the screen.

**Pitch (cam 13)**, tempo, and volume may be **altered (cam 9)** to **indicate (cam 10)** how the filmmaker expects the **audience (cam 11)** to respond to a given noise. For instance, high-pitched sounds, including screams or squealing tires, help to create a sense of anxiety, while low-pitched sounds, including the sounds of waves or the swinging of a door, can be used to create a sense of calm or **mystery (cam 12)**.

Perhaps the most interesting use of sound in a movie is the very **absence (cam 12)** of it: silence. At key points in a film, directors may use silence in much the same way that they would use a freeze **frame (cam 12)**. Both tend to **arrest**

**refers to** = cite = mention = name

**scene** = a part of a play or film in which the action stays in one place for a continuous period of time

**Plot**= story of film/book...

**Category**= a category in retailing is a group of the same products (ex: soft drinks, detergents, etc)

**Element**= factor.

**fictional** = imaginary

**Atmosphere**= air in environment

**Pitch= tone** = how high or low a sound is

**Altered**= changed # original

**Indicate**= point out, show, suggest...

**audience** = the group of people together in one place to watch or listen to a play, film,...

**Mystery**= secrecy, unknown, unidentified...

**absence**= lack, non-existence# presence

**Frame**= structure

**Arrest** = capture = seizure # release

**(arrested –cam 12, arresting –cam 13)** the audience's attention to highlight some action or change in story direction. Silence can be used to build up a scene's **intensity (cam 9)** or to **foreshadow (cam 11)** impending **doom (doomed – cam 11)**

In recent years, special sound effects have been added to movies in order to heighten the film experience. Many of these sound effects, including **explosions (cam 11)**, phaser **blasts (blasted –cam 12,13)**, wind, and animal sounds are drawn from computer sound effects libraries and are added to a film after the movie has been shot. Besides creating louder and more dramatic movies, these effects have tended to draw more attention to movie sound. With advancements in surround sound, sound effects have developed a more "directional"

element, appearing to come from a specific place or direction. This directional quality of sound (alongside elements such as echoes) **enhances (cam 12)** a **three-dimensional (dimension – cam 12)** sense of space in the movie.

How Does Music Help to Shape a Film?

If we step back and think about it, music is one of the most **peculiar (Cam 10) conventions (cam 11)** in movies. No one questions that music should be a part of movies because we've all grown used to the idea that, in a movie, when two people kiss, we should hear music in the background. Or when the platoon attacks the beach, a **symphony (cam 12)** should provide the **inspiration (cam 12)** behind their assault. Of course, no one has a soundtrack **accompanying (cam 9,12)** their real lives. But in movies we not only accept this convention, we demand it.

Music can be used for a number of effects in a movie. The most obvious way music scores are used is to guide the emotional response of the audience. They provide **clues (cam 8,10)**, or, in most cases, huge signposts, that tell audiences how the filmmaker wants them to react to a given scene.

**Intensity**= strength, power, amount...

**Foreshadow**= anticipate

**Doom**= ruin, destroy

**Explosion**= bang

**Blast**= criticize

**Enhance**= improve, boost, enrich, increase

**three-dimensional** = having or appearing to have height, length, and width and therefore looking real

**peculiar**= strange = weird = unusual

**Convention**= standard, rule, principle

**Symphony**= a long piece of music usually in four parts, written for an orchestra.

**Inspiration**= a person, experience, place etc that gives you new ideas for something you do

**Accompany**= go along with, go together with...

**Clue**= evidence, sign...

Some directors play against our expectations and use music in ways we might not expect. Stanley Kubrick shocked audiences when he used “Singin’ in the Rain” as the backdrop to a horrible rape scene in *A Clockwork Orange* (1971). Music can also provide an overture for a movie when it’s used as the **backdrop (cam 9)** for the opening **credits (cam 11)**. The brassy **theme (cam 9)** music **composed (cam 13)** by John Williams for Star Wars is one famous and often-parodied example. In some instances, directors use music to **foreshadow** upcoming events. In horror movies, for example, the score is often used to build up **tension (cam 8)** and **suspense (cam 11)** just before the monster attacks one of its victims. Finally, music can be used to shape the **ethnic (cam 13)** or cultural context of a film.

**Backdrop**= the conditions or situation in which something happens.

**credits** = the list of the names of people and organizations who helped to make a movie

**Theme**= **subject**, topic, subject matter

**Composed by** = written by = created by

**Tension**= worry, nervousness, anxiety...

**Suspense**= uncertainty, doubt, anxiety, nervousness

**ethnic** = relating to a race or national group of people

### PASSAGE 3: THIS MARVELLOUS INVENTION

#### ARTICLE: Top 10: Life's greatest inventions

##### 4. Language

As far as humans are concerned, language has got to be the **ultimate (cam 9) evolutionary (cam 8,10) innovation (cam 9)**. It is central to most of what makes us special, from **consciousness (conscious –cam 12), empathy (cam 9)** and mental time travel to symbolism, **spirituality (spirit –cam 8)** and **morality (cam 12)**. Language may be a **defining (cam 8)** factor of our species, but just how important is it in the evolutionary **scheme (cam 11)** of things?

A decade ago, John Maynard Smith, then emeritus professor of biology at the University of Sussex, UK, and Eors Szathmary from the Institute of Advanced Study in Budapest, Hungary, published The Major Transitions in Evolution, their description of life's great leaps forward. They identified these crucial steps as innovations in the way information was organised and **transmitted (cam 8)** from one generation to the next – starting with the origin of life itself and ending with language.

Exactly how our **ancestors (cam 9)** took this leap is possibly the hardest problem in science, Szathmary says. He points out that **complex (Cam 12,13)** language – language with syntax and grammar, which builds up meaning through a hierarchical arrangement of subordinate clauses – **evolved (cam 9)** just once. Only human brains are able to produce language, and, contrary to popular belief, this ability is not **confined (cam 8)** to specialised regions in the brain such as Broca's and Wernicke's areas. If these are damaged

**Ultimate** = crucial = vital = decisive

**Evolutionary**= relating to the way in which plants and animals develop and change gradually over a long period of time.

**Innovation**= modernization, improvement, advance...

**Consciousness**= awareness = realization = perception

**Empathy**=the ability to understand other people's feelings and problems .

**spirituality** = Deep feelings and beliefs of a religious nature

**Morality**= ethics.

**Define**= describe correctly and thoroughly

**Scheme**= **plan**, project, programme, strategy

**Transmit**= convey, pass on...

**Ancestor**= a person related to you who lived a long time ago

**complex** = complicated, Intricate ...

**Evolve**= develop, change, grow...

**Confine** = restrict = limit = kept

others can take over. Szathmary likens language to an amoeba, and the human brain to the **habitat (cam 12)** in which it can **thrive (cam 11)**. “A surprisingly large part of our brain can **sustain (sustainable –cam 10,11)** language,” he says.

But that raises the question of why this language amoeba doesn’t **colonise (colonized – cam 12)** the brains of other animals, especially primates. Szathmary is **convinced (cam 8,10)** the answer lies in neural networks unique to humans that allow us to perform the complex hierarchical processing required for grammatical language. These networks are shaped both by our genes and by experience. The first gene associated with language, FOXP2, was identified in 2001, and others will surely follow.

So why don’t our close evolutionary relatives, **chimps (cam 11)** and other primates, have similar abilities? The answer, recent analysis seems to suggest, lies in the fact that while humans and chimps have many genes in common, the versions expressed in human brains are more active than those in chimps. What’s more, the brains of newborn humans are far less developed than those of newborn chimps, which means that our neural networks are shaped over many years of development **immersed (cam 9)** in a linguistic environment.

In a sense, language is the last word in biological evolution. That’s because this particular evolutionary innovation allows those who possess it to move beyond the realms of the **purely (cam 11)** biological. With language, our ancestors were able to create their own environment – we now call it culture – and **adapt (cam 8)** to it without the need for genetic changes.

**Habitat**= the natural home of a plant or animal.

**Thrive**= flourish = Grow well

**sustain**= support = maintain

**colonise** = to establish political control over an area, and send your citizens there to settle

**Convinced**= persuaded.

**Chimp** = an intelligent African animal that is like a large monkey without a tail

**Immersed in** = completely involved in an activity

**Purely**= completely and only

**Adapt**= get a feel to, get used to, familiarize yourself...